

Analysis of time-to-event for observational studies: Guidance to the use of intensity models

Per Kragh Andersen¹ | Maja Pohar Perme² | Hans C. van Houwelingen³ | Richard J. Cook⁴ | Pierre Joly⁵ | Torben Martinussen¹ | Jeremy M. G. Taylor⁶ | Michal Abrahamowicz⁷ | Terry M. Therneau⁸

¹Section of Biostatistics, University of Copenhagen, Copenhagen, Denmark

²Department of Biostatistics and Medical Informatics, Medical faculty, University of Ljubljana, Ljubljana, Slovenia

³Department of Biomedical Data Sciences, Leiden University, Leiden, The Netherlands

⁴Department of Statistics and Actuarial Science, University of Waterloo, Waterloo, Ontario, Canada

⁵Inserm, ISPED, Bordeaux Populations Health Research Center, University of Bordeaux, Bordeaux, France

⁶Department of Biostatistics, University of Michigan, Ann Arbor, Michigan, USA

⁷Department of Epidemiology, Biostatistics and Occupational Health, McGill University, Montreal, Quebec, Canada

⁸Division of Biomedical Statistics and Informatics, Mayo Clinic, Rochester, New York, USA

Correspondence

Maja Pohar Perme, Department of Biostatistics and Medical Informatics, Medical Faculty, University of Ljubljana, Ljubljana, Slovenia.
Email: maja.pohar@mf.uni-lj.si

Funding information

Canadian Institutes of Health Research, Grant/Award Number: PJT-148946; Javna Agencija za Raziskovalno Dejavnost RS, Grant/Award Number: P3-0154; Natural Sciences and Engineering Research Council of Canada, Grant/Award Number: 228203

Summary

This paper provides guidance for researchers with some mathematical background on the conduct of time-to-event analysis in observational studies based on intensity (hazard) models. Discussions of basic concepts like time axis, event definition and censoring are given. Hazard models are introduced, with special emphasis on the Cox proportional hazards regression model. We provide check lists that may be useful both when fitting the model and assessing its goodness of fit and when interpreting the results. Special attention is paid to how to avoid problems with immortal time bias by introducing time-dependent covariates. We discuss prediction based on hazard models and difficulties when attempting to draw proper causal conclusions from such models. Finally, we present a series of examples where the methods and check lists are exemplified. Computational details and implementation using the freely available R software are documented in Supplementary Material. The paper was prepared as part of the STRATOS initiative.

KEYWORDS

censoring, Cox regression model, hazard function, immortal time bias, multistate model, prediction, STRATOS initiative, survival analysis, time-dependent covariates

1 | INTRODUCTION

Methods for survival, or time-to-event, analysis are frequently used in epidemiological and clinical studies of human health. The more than 30 000 Pubmed citations for the Cox proportional hazards (PH) model alone attest to the critical role of such methods in modern health research. Most of the observable health outcomes, such as disease onset, progression, cure or death, are the result of the evolution of relevant biological systems resulting from a natural aging process or the effects of exposures and treatments that may accumulate over time; hence a time-to-event paradigm provides a natural framework for their analyses. Accordingly, biostatisticians working in medical research are very likely to encounter problems requiring time-to-event analyses, even if their training and interests lie in different areas of statistical research. Time-to-event data typically feature particular challenges related to, among other things, censored observations and changes over time in the absolute and/or relative risks, as well as in the values of the predictors. To further complicate matters, there are several issues commonly arising in survival analysis which are poorly understood, for which there are no guidelines, or for which there is no consensus on how to best handle them. Examples include immortal time bias, time-dependent variables, censoring vs competing risks, and subtleties in the interpretation of model results. The lack of clear guidance on how to address these challenges may explain why many published applications involving survival analysis are found to have important weaknesses, see for example, Altman et al.¹

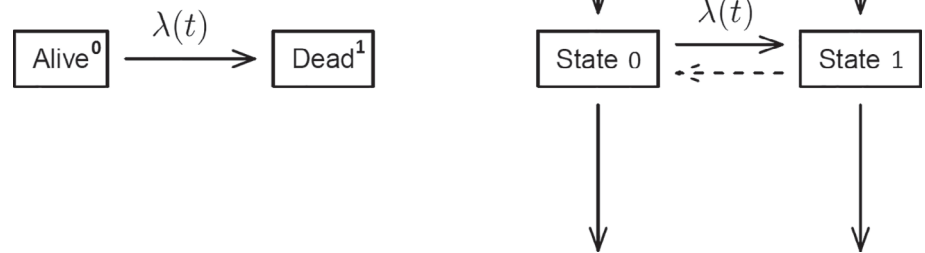
These considerations motivated us to create the “Survival Analysis” topic group within the STRATOS (STRengthening Analytical Thinking for Observational Studies) initiative. The over-arching aim of the STRATOS initiative is to provide guidance for accurate and efficient analyses in different areas of statistics relevant for observational (nonrandomized) studies.² The current paper reflects the discussions within our STRATOS topic group (TG8), and presents the first step toward a coherent approach to real-life applications of survival analyses based on intensity (or “hazard”) models. In particular, we discuss fundamental assumptions, outline the basic steps necessary to ensure that the analysis appropriately uses the data at hand to address the substantive research question. We also discuss some pitfalls and ways to avoid them, point out some subtle complexities that may arise in applications, and suggest how the basic methodology may be adapted or extended to address these additional issues.

In many observational cohort studies, interest lies in the occurrence of a particular *event* among subjects with a given condition (ie, those “exposed”) and among those without the condition (“unexposed”), and the goal may be to compare the pattern of event occurrence. In other settings factors of interest may evolve over time as exposure changes with varying treatments. Consider a register study of the association between exposure to the drug lithium and the incidence of dementia³ (later referred to as the lithium and dementia study). In this study, the event is a hospital diagnosis of dementia and the levels of exposure correspond to different numbers of redeemed prescriptions (0, 1, 2, ...) of the drug lithium. The lithium and dementia study will be used to set examples throughout the first sections of the paper while other studies and datasets will be used for illustrations in Section 6. There, for example, we study the risk factors for all cause mortality in women with ovarian cancer and discuss more complex clinical cohorts and raise the issue of what can (and cannot) be reliably estimated from the samples in the studies of patients with nonalcoholic fatty liver disease (NAFLD) and peripheral arterial disease (PAD).

The common situation for the examples is shown in Figure 1. In such settings there is a time axis usually measuring the time from the origin of the process or some other natural starting point that may be defined by the context. We denote this by t and presume it is measured on a common scale for all individuals of interest. The left-side graph refers to the simplest situation, where death of any cause is the only event of interest as in the ovarian cancer study. There the time origin would most naturally be the age of diagnosis with ovarian cancer. The subjects in the *state* denoted as “0” are alive (with ovarian cancer) and, thereby, *at risk* of experiencing the event and the state corresponding to this event is denoted “1.” The right-side graph refers to a more general situation, where the event of interest may be part of a larger system containing aspects that may or may not be relevant for the study. In that case, the event may occur via other states than “0,” other events (transitions) may happen to subjects in state 0, thereby preventing the occurrence of the event (competing risks), and subjects may or may not return to state 0 after the occurrence of the event (dashed line). For subjects in state 0, there is a probabilistic *intensity*, say $\lambda(t)$ (or $\lambda_{01}(t)$), governing event occurrence and this intensity is often the primary target for statistical modelling.

An important point to note in connection with such studies is that the event of interest will typically not be observed for all subjects. This incomplete information could be caused by different mechanisms, including being event-free at end of follow-up (ie, still in state 0 when the study ends or at an earlier time due to loss of follow-up) or experiencing a competing event (ie, leaving state 0 to another state than 1). We will discuss the corresponding concepts of *censoring* and *competing risks* in detail in what follows.

FIGURE 1 Representation of multistate processes and transitions between states. On the left is the simplest case of survival until death from any cause, on the right a more general situation where the transition of interest (from state 0 to state 1) is part of a larger multistate model



We discuss survival analysis using intensity models on data from cohort studies like those described in the examples above. We will emphasize that there may be different types of scientific questions to be addressed in such observational cohort studies, and that the analysis should be properly targeted to those questions. Nonetheless there are a number of special features of survival data arising from such studies of which investigators should be aware. In Section 2, we will discuss such features with examples and give recommendations. We will also discuss potential pitfalls connected with such analyses and how to avoid them. In Section 3 we will focus on models for the intensity $\lambda(t)$. We will see that, for such an analysis, a more detailed description of other (“competing”) events that subjects may experience while being at risk for the event of interest may not be needed. However, an important point will be that even though such intensity models may suffice for addressing some questions, one will have to go beyond intensities to deal with other questions, including estimation of the absolute risk of experiencing the event (Section 4) and also various *causal* questions (Section 5). This is the focus of Sections 4 and 5. Section 6 illustrates the relevant issues and methods through analysis of the above examples, and the paper is concluded by a brief discussion in Section 7.

2 | GETTING THE BASICS READY

This section introduces the notation to be used throughout the paper and defines the intensity. We also give a checklist of items that are important to consider in observational time-to-event studies.

2.1 | Notation

We assume that the following data can be available for subject i in a sample of n independent individuals, $i = 1, \dots, n$:

- The follow-up time T_i , that is, the time from the chosen time origin at which observation of the subject ended. (The uncensored, incompletely observed time to death is denoted by T_i^*).
- The indicator variable δ_i , indicating whether or not, at time T_i , the event of interest occurred ($\delta_i = 1$ for event, ie, $T_i = T_i^*$, and 0 otherwise).
- A time V_i (relative to the chosen time origin where $V_i < T_i$) where the subject entered the study. Thus, if the subject was included already at the chosen time origin then $V_i = 0$, but $V_i > 0$ (*delayed entry*) is possible.
- A vector of covariate values $Z_i(t) = (Z_{i1}(t), \dots, Z_{ip}(t))$; $t \in [V_i, T_i]$, some of which may be time-dependent but other may be time-fixed (baseline).

Two common approaches exist for notation in survival data. A traditional way to describe the survival data is the vector $(T_i, V_i, \delta_i, Z_i(t))$, and this works well for time to all-cause death, the simplest classic example of survival data (Figure 1, left panel). Alternatively, one can view each subject as an evolution over time leading to the counting process notation which accommodates more complex situations such as the multistate process implied by the right diagram of Figure 1; it will be hence used throughout the paper. Here, the process $Y_i(t)$ is equal to 1 while a person is known to be at risk and 0 otherwise (ie, $Y_i(t) = I(V_i < t \leq T_i)$) and the process $N_i(t)$ denotes the number of observed events by time t , here

simply given by $N_i(t) = I(V_i < T_i \leq t, \delta_i=1)$. Defining $dN_i(t) = N_i(t) - N_i(t-)$, observation of the event at time t can then be represented by $dN_i(t) = 1$, that is, the counting process for individual i increments by 1 at that time.

2.2 | Preliminary concepts and issues

The most important aspect of any analysis is to first think carefully about (i) what question(s) we want to answer, and (ii) whether and how the data at hand will be sufficient to answer them. With respect to the latter (ii), important issues are the source of the data, what population it represents, what variables are relevant and which among these are available, and data completeness, both with respect to inclusion of subjects and missing data for those that are included. We can then proceed with a more technical checklist:

- **Time origin:** The follow-up time T_i is measured from a meaningful starting point of the process ($t = 0$), which should be unambiguously defined, comparable between individuals, and ideally clinically relevant. Typical examples include age, time since diagnosis or time since treatment initiation. The choice of time origin should depend on the scientific questions.

In the lithium and dementia study, the time origin was defined as the date of the start of the first lithium prescription for a given patient. Patients who started lithium treatment after 1996 are included in the study. In the PAD study, the patients with peripheral artery disease were identified at a visit to their physician. While the onset of the symptoms may be observable, it is impossible to know their times of the disease onset.

The time origin does not always correspond to any “clinically meaningful” date but to an administrative date of the “start of the follow-up,” so the study results must be interpreted accordingly. In these cases age (as a major determinant of many health outcomes) may be the appropriate time axis, and it is increasingly used in population based studies investigating development of diseases. If the underlying risks change systematically with, for example, time since diagnosis, then this will be the preferable time axis.

The definition of the time origin determines the primary time axis. In the lithium and dementia study, the time of interest is the time from treatment initiation to dementia onset. As the time origin is the time of lithium treatment initiation, all patients are at risk at time $t = 0$, that is, $V_i = 0$ and $Y_i(0) = 1$ for all i . The study also addressed a second research question comparing the lithium treated patients to the general population and a sample from the general population is followed from 1st of January 1995 onward. In the general population, the time since lithium initiation of course cannot be defined and, thus, to answer the second research question, one should then use the age as the time axis instead, for both the lithium treated and the general population subjects. This is an example of delayed entry for all individuals, everyone has $Y_i(0) = 0$, since no one is included into the study at birth, the $Y_i(t)$ values switch to 1 at the age at which the individual was included in the study. Accounting for the delayed entry is necessary here to avoid so-called *immortal time bias*,^{4,5} see Section 3.4.

To address the important question in the study *multiple time axes* are needed—both time since lithium treatment initiation and age. This is a common feature in epidemiological follow-up studies.

- **Inclusion criteria:** The inclusion criteria for individual i must be met by the time that the patient is declared to enter the study, that is, at time V_i when $Y_i(t)$ first becomes 1.

Suppose that we wished to analyze survival time of the general population in the lithium and dementia study. One cannot say that only individuals who are never treated with lithium throughout the study can be used for this analysis. While this information may be known later, at the date of data analysis, it was not known when the individuals were included.

If the individuals who were treated with lithium at a certain point later in the study were excluded from the study, this would imply that patients with a shorter time to event (and hence less time to get lithium treatment) were more likely to be included and the survival of this general population group would be underestimated.

On the other hand, if “ever treated” individuals from the general population were included and were considered to be a part of the patient group throughout the study, this would in turn imply the overestimation of patients’ survival. This is another example of immortal time bias. A correct way of analysis of these data is to regard the exposure (lithium treatment) as a time-varying covariate, we return to this in Section 3.4.

- **Event definition:** The time of event occurrence must be clearly defined. In the PAD study several events are of interest. The time of death or the time of a major cardiovascular (CV) event (stroke, infarction) are examples of well-defined events with an exact date typically known. Other types of events, such as revascularization procedure are slightly more

subjective in nature, since this is an operation that has been scheduled following the doctor's evaluation of the patient's state of the disease. Judgement is therefore involved and moreover, the date when it took place depended not only on the stage of the disease but also the ability to schedule the procedure. Another example of an event with some ambiguity is the onset of dementia. Here, one knows that the onset of dementia has happened between two consecutive visits, but it is impossible to know when. This means that the onset of dementia is interval censored. This could be further complicated if not all patient visits were scheduled with the same frequency, whereby some patients may be diagnosed earlier than others. In the lithium exposure study, these problems were avoided by *defining* the event as the first hospitalization leading to a diagnosis of dementia (which may not coincide with the actual onset of the disease and must thus be interpreted accordingly).

The decision whether the event of interest has already occurred must be known at the time when $N(t)$ switches to 1. A typical example occurs with validated endpoints. An endpoint such as diabetes might be miscoded, for instance, and investigators will often require “two diagnoses at least 30 days apart” as proof. An error occurs when the date of diabetes is backdated to be that of the first instance and, therefore, such an event definition would depend on something happening in the future.

- **Censoring:** The goal of the survival analysis is to estimate quantities relating to a complete, that is, uncensored, population. A basic assumption in the estimation is that the information that a patient has *not been censored* at a certain time point does not carry any information about his or her prognosis beyond that time point. We need this assumption since we shall regard the patients at risk at a certain point t ($Y_i(t) = 1$) as a representative subsample of all the patients that would be at risk if there was no censoring. The assumption is referred to as *independent censoring*.

The assumption can be weakened to *conditionally* independent censoring, i.e. independent censoring within a group of patients with a certain set of characteristics, defined by the covariates available in the study data base. Administrative censoring, that is, censoring at a certain calendar time due to the end of study, is a common example of independent censoring, as the censoring mechanism is not related to individual patient prognosis. However, if the patient prognosis has improved through the calendar time covered by the study, the patients diagnosed later have a better prognosis and are also censored earlier, so the independent censoring assumption is not met. On the other hand, the censoring pattern is conditionally independent given the period of diagnosis, so inclusion of this covariate in the model will avoid potential bias (provided the model is correct). It is, thus, important to consider what causes censoring in any given follow-up study. Is it mostly administrative censoring, that is, being event-free at end of planned follow-up, or are there drop-outs? While the former can often be taken to be independent, more suspicion should be exercised for the latter and, ideally, it should be noted in the data set *why* any given subject was censored.

In some studies one considers deaths due to causes unrelated to the disease in question as censoring, for example, regarding the patients who die from non-CV causes in the PAD study (causes that were not of the main interest in this study) to be equivalent to those who were administratively censored. This is not a formal violation of the definition of independent censoring, rather it is inconsistent with the definition of our population of interest. This is because the competing risk (death from non-CV causes) is present also in the complete population and we usually are not interested in the population where this risk would be eliminated. We address this situation in the Competing risks section (Section 4.2).

- **Delayed entry:** Similar to this discussion, the time V of delayed entry should also be “independent” in the sense that knowing at time $t > V$ that a subject entered the study at V should not carry information about his/her prognosis beyond t .⁶

2.3 | The intensity

We now return to the concept of the *intensity* function. Once it has been established who is at risk, what is the event and how should subjects should be aligned over time (ie, how time t is defined), one can define the *intensity* for subject i as:

$$\begin{aligned}\lambda_i(t) &\approx P(\text{event in } (t, t + dt) | \text{past at time } t-) / dt \\ &= P(dN_i(t) = 1 | H_i(t-)) / dt,\end{aligned}\quad (1)$$

where $H_i(t-) = (N_i(s), Y_i(s), Z_i(s), s < t)$ summarizes the past information for the subject i that is available just before time t . (More formally: $\lambda_i(t) = \lim_{\Delta t \rightarrow 0} P(dN_i(t) = 1 | H_i(t-)) / \Delta t$.)

For *survival data* involving only states 0 and 1 as in the left panel of Figure 1, the *hazard function* is given by:

$$\lambda(t) = -\frac{d \log S(t)}{dt}, \quad (2)$$

where the *survival function* $S(t)$ is $P(T^* > t)$. When no time-dependent covariates are considered in $H_i(t)$, the intensity in (1) is simply given by the hazard in (2) and, for that reason, we will use the terms *intensity* and *hazard* interchangeably in this paper. While the hazard function for survival data (as seen in equation (2)) is in one-to-one correspondence with $S(t)$, and thereby with the cumulative risk $F(t) = 1 - S(t)$, there may more generally be other events competing with the event of interest in which case $\lambda(t)$ is a transition hazard (or cause-specific hazard) from state 0 to state 1 (and it is some times denoted $\lambda_{01}(t)$ to emphasize this). In that case, a one-to-one correspondence between the single cause-specific hazard and the absolute probabilities no longer exists, see, for example, Andersen et al.⁷

There are several reasons why the intensity function plays a central role in survival analysis and analysis of cohort studies.

- The idea is that subjects are followed over time and, at each time t where the subject is still observed to be at risk, it is asked, given the information that is available so far for the subject, what is then the probability per time unit that the subject experiences the event in the next little time interval from t to $t + dt$? Thus, the intensity gives a dynamical description of how events occur over time and, in this description, all aspects of the *past* observed for the subject up to time t may be taken into account, such as (time-dependent) covariates and, possibly, previous events.
- For survival data, the survival probability $S(t) = P(T^* > t)$ cannot be estimated in a straightforward way as a simple proportion due to censoring. The hazard function, however, can still be studied since it relies exclusively on the information on the individuals still at risk (assuming the independent censoring assumption to hold).
- One of the greatest strengths of intensity-based models is that one can examine the effects of both *baseline covariates* (ie, information available at time of entry into the study) and *time-dependent covariates* that are updated during follow-up. Time-dependent covariates have a very natural connection to clinical practice: when assessing a long-term patient, a physician will naturally use the most recent measurements in addition to those collected a long time ago at their initial visit. Time-dependent covariates thus allow accounting for the changes over time in the relevant variables (patient characteristics, exposures, treatment), which may alter the intensity.

In the lithium and dementia study, the exposure itself (number of lithium prescriptions redeemed until the current time) is time-dependent and, when including the unexposed control group of nontreated subjects, people will change status from being unexposed to belonging to the exposed group at the time of their first lithium prescription.

Unfortunately, misunderstandings and errors in creating time-dependent covariates are one of the most common sources of immortal time bias, we thus pay this issue special attention in Section 3.4. Time-dependent covariates also create issues with model prediction, which will be discussed in Section 4.1.

- As the intensity is a dynamic description of the data generating process over time, delayed entry is naturally taken into account.
- Most often, the hazard depends on patient characteristics, so that *covariates* need to be taken into account when analyzing data from cohort studies. In this context, as discussed in Section 2.2 above, an advantage of using a hazard regression model is that it “corrects for non-independent censoring” in the sense that regression coefficients are estimated consistently even if censoring depends on covariates, as long as these covariates are included as predictors in the hazard model (eg, Andersen et al,⁸ section III.2).

We argue that the intensity may be of interest in its own right and that it, therefore, could be an obvious target for an analysis of the cohort data. However, as previously mentioned, there are important scientific questions for the answer of which the intensity will be insufficient. We will return to that in Sections 4 and 5.

3 | HAZARD MODELS

In order to make a *model* for the intensity one needs to specify how it depends on time t and on the available information in $H_i(t)$, more specifically: how it depends on the covariates.

Before specifying a hazard model, descriptive analysis should be conducted to explore the data. The *Kaplan-Meier estimator* for the whole cohort, or for subgroups, will provide useful insights in the case of a single (possibly composite) terminal event, see the left panel of Figure 1. However, this will not be the case in, for example, the lithium and dementia study where, obviously, all-cause mortality is a competing risk for dementia, the event of interest. To describe the *hazard*, one could divide the time variable of interest into suitable intervals (eg, yearly intervals) and calculate the *incidence* (or “occurrence-exposure”) rate for each interval by dividing the total number of events of interest observed in the interval by the total time at risk in the interval. This corresponds to an assumption of a piecewise constant hazard function, an assumption that is often a reasonable approximation and which is used for *Poisson regression* which will be discussed further below.

To estimate the hazard *nonparametrically* requires some sort of smoothing. What may be estimated in a simple non-parametric fashion for a defined population of subjects is the *cumulative hazard* $\Lambda(t) = \int_0^t \lambda(u)du$. This may be done using the *Nelson-Aalen estimator*

$$\hat{\Lambda}(t) = \int_0^t \frac{dN(u)}{Y(u)} = \sum_{i: T_i \leq t} \frac{\delta_i}{Y(T_i)}, \quad (3)$$

where $Y(t) = \sum_i Y_i(t)$ and $N(t) = \sum_i N_i(t)$. This is an increasing step function with steps at each observed event time, the step size at t being inversely proportional to the number $Y(t)$ at risk, Figure 8 shows an example. Pointwise confidence limits may be added (eg, Andersen et al,⁸ chapter IV). The approximate “local slope” of $\hat{\Lambda}$ at t reflects the hazard at that time. However, the value of the cumulative hazard, does, in general, not have a simple interpretation (an exception is in studies of *recurrent events*⁹), a topic beyond the scope of this paper.

3.1 | PH models

Most applications of intensity models for cohort studies aim at relating the rate of event occurrence to (time-fixed or time-dependent) *covariates*. The complexity of such a hazard model will both depend on what is the scientific question that it is meant address and on what information is available. Here, it is important to notice that hazard models, as any other statistical model, can typically not be expected to be “correct” in any strict sense but still they may be sufficiently flexible to give a sensible answer to the question raised.

Survival analysis and, thereby, also analysis of cohort studies is dominated by the Cox model. In the breakthrough paper,¹⁰ the maximum partial likelihood was introduced as a method to estimate the regression coefficients in the PH model. The general definition of the PH model is

$$\lambda_i(t) = \lambda(t|Z_i(t)) = \lambda_0(t) \exp(Z_i(t)^\top \beta). \quad (4)$$

In the Cox PH model, the *baseline hazard* $\lambda_0(t)$ (ie, the hazard for individuals with all covariates equal to 0) is left unspecified, other alternatives for the PH model are considered at the end of this section. The name “proportional hazards” refers to the, possibly strong, assumption that the ratio of the hazards corresponding to two different values of $Z_i(t)$ is the same for all times t . This constant hazard ratio is $\exp(\beta)$. In this model, the regression parameter(s) β are estimated by maximizing the log partial likelihood

$$pl(\beta) = \sum_{i=1}^n \int_0^\infty \log \left(\frac{\exp(Z_i(t)^\top \beta)}{\sum_j Y_j(t) \exp(Z_j(t)^\top \beta)} \right) dN_i(t). \quad (5)$$

Once the covariate effects β have been estimated, the cumulative baseline hazard $\Lambda_0(t) = \int_0^t \lambda_0(u)du$ can be estimated by the Breslow estimator¹¹

$$\hat{\Lambda}_0(t|\hat{\beta}) = \int_0^t \frac{dN(u)}{\sum_i Y_i(u) \exp(Z_i(u)^\top \hat{\beta})}. \quad (6)$$

With no covariates ($\beta = 0$) the Breslow estimator is simply the Nelson-Aalen estimator (3). The partial likelihood may be seen as a profile likelihood resulting from eliminating the baseline hazard from a joint likelihood including both β

and $\lambda_0(t)$. It is important to notice that this joint likelihood is valid both for all-cause survival data (left panel of Figure 1) and for the more general situation depicted in the right panel of that figure. This is because the likelihood for the whole system *factorizes* into a number of factors, each depending on a separate transition in the model.^{8,12} Also, the Cox partial likelihood (5) enjoys the properties of standard likelihood functions such that standard errors and test statistics may be obtained in the “usual way.”^{13,14} We will use these features in the Examples Section 6.

If more time axes are relevant in a given study then, using the Cox model, one of these must be selected as the baseline time axis t . Other time axes (eg, current age if t is time since disease onset) can then be included as time-dependent covariates. This type of time-dependent covariates is said to be *external* (or *exogenous*) because they “exist” whether or not the subject is still under observation. On the other hand, time-dependent covariates such as current blood pressure or current cholesterol level (that can only be ascertained for subjects still under observation) are *internal* (or *endogenous*).

The completely unspecified baseline in the Cox model makes it quite flexible, however, a limitation of this non-parametric model component is that it only allows direct estimation of the *cumulative* baseline hazard $\Lambda_0(t)$, but fails to produce an estimate of the hazard $\lambda_0(t)$ itself. To obtain an estimate of $\lambda_0(t)$, some smoothing would be required.

The alternative to letting the baseline hazard remain unspecified in the model is to fit a fully parametric PH model, some of the common options are:

- The simplest (and most restrictive) option is to assume a *constant* baseline hazard corresponding to an exponential distribution of T^* in the case of all-cause survival data.
- A useful extension of the model above is the *piecewise exponential* model that divides the time-range into intervals on which the baseline hazard is constant. Here, cut-points $0 = s_0 < s_1 < \dots < s_{K-1} < s_K = \infty$ for the time axis are selected and it is assumed that $\lambda_0(t) = \lambda_{j0}$ when $s_{j-1} \leq t < s_j$. This is often referred to as the *Poisson* (or piecewise exponential) regression model for survival data and it is frequently used in epidemiological studies. It tends to produce results that are very close to those obtained using the Cox model. An advantage is that the baseline hazard is flexible despite being fully parametric. Further advantages are that a potentially large (eg, registry-based) dataset can be preprocessed into tables of *event counts* and *person-time at risk* according to the chosen time intervals and to (discrete-valued) covariates and, furthermore, that *multiple time axes* are very easily handled by splitting event counts and person-years at risk simultaneously according to all time axes.¹⁵ Drawbacks of the piecewise exponential model include the fact that the intervals must be selected and that this choice to some extent may affect the detailed results and, furthermore, that it does not produce a *smooth* hazard function.
- A smooth extension of the constant baseline hazard model is the Weibull model, where $\lambda_0(t) = \lambda\gamma(\lambda t)^{\gamma-1}$. The extra parameter γ allows some flexibility, but assumes a monotone baseline hazard function and the model is not flexible around $t = 0$. To allow greater flexibility and obtain a smooth baseline hazard one may use flexible parametric models for $\lambda_0(t)$, for example, via splines in combination with penalized likelihood.¹⁶

3.2 | Alternatives to PH models

The PH assumption is strong and may often not fit the data well for the entire time range studied. An extension of the Cox model, to relax the PH assumption, is to allow the covariates $Z(t)$ to have *time-varying* effects, that is, assume that the hazard is given by

$$\lambda(t|Z_i(t)) = \lambda_0(t) \exp(Z_i(t)^\top \beta(t)). \quad (7)$$

Here, explicit interactions between covariates and functions of time may be introduced, for example, by defining a model with $\beta(t) = \sum_j \gamma_j f_j(t)$ (for a set of prespecified functions $f_j(t)$ containing $f_0(t) \equiv 1$) for each component of Z . The simplest example is to split time into two intervals (splitting at time τ , say) and assume PH within each. This corresponds to choosing $f_0(t) = 1$ and $f_1(t) = I(t \geq \tau)$. Alternatives include the use of splines.¹⁷ Some care is needed here since the number of parameters in such models with time-varying effects can become quite large and there is a danger of overfitting. In the situation where the PH assumption needs to be relaxed for a single categorical covariate, the *stratified Cox model* is useful. In this model, each level of that covariate has its own baseline hazard which is not further specified, that is,

$$\lambda(t|Z_i(t)) = \lambda_{0s}(t) \exp(Z_i(t)^\top \beta), \quad (8)$$

when subject i belongs to stratum s .

An alternative to the multiplicative PH models is an *additive hazards model*. In the *Aalen model*¹⁸

$$\lambda(t|Z_i(t)) = \lambda_0(t) + Z_i(t)^\top \beta(t), \quad (9)$$

both the baseline hazard $\lambda_0(t)$ and the regression functions $\beta(t)$ are completely unspecified (like the baseline hazard for the Cox model) and their cumulatives $\int_0^t \lambda_0(u)du$, $\int_0^t \beta(u)du$ can be estimated using a least squares technique. Versions of the model where some or all $\beta(t)$ are time-constant are also available.^{19,20} A drawback of the Aalen model is that the estimated hazard can become negative while an advantage is that it is very flexible.²¹

A completely different approach is given by the accelerated failure time (AFT) model where the covariates are assumed to extend or shorten the survival time by a constant time ratio $\exp(\beta)$ e.g. Kalbfleisch and Prentice, Chapter 7¹²

$$S(t|Z) = S_0(\exp(-Z^\top \beta)t),$$

or equivalently:

$$\ln(T^*) = Z^\top \beta + \epsilon.$$

The model is a viable alternative to the PH model and although one could derive the hazard function for this model, it does not naturally fall under the heading of “hazard models.” Also, it is mostly used for survival data (Figure 1, left part) and less so in the general situation of Figure 1, right part, and it will not be considered further in this paper. Discussion of pros and cons of PH and AFT can be found in References 22 and 23.

3.3 | A checklist when fitting the Cox model

We next propose a checklist for the Cox model. Most of the items below are relevant also for other hazard regression models. We list the issues that one should be careful about both before fitting a model and after having performed an analysis. We add tests and approaches that can be helpful in understanding the sources of the problems and evaluating their extent. Note, however, that these checks are not conclusive, they serve only as an aid in thinking about the issues.

Before fitting a model the following items should be considered:

- *Checks on the covariates* to be included in the model: for continuous covariates examine the distribution, check for extreme data (leverage) points, make histograms. For categorical covariates, the frequencies of the categories should be reported and also the choice of the reference categories.
- *Check dates*. A trivial, but often relevant warning is that survival data often contain a series of dates, that may come in varying formats and are prone to typing mistakes. The fact that the dates follow each other in the proper sequence should thus be carefully checked.
- *Investigate censoring*. As mentioned in the previous checklist, it is first of all important to think about what causes censoring. Next, plotting a “survival curve” estimating $C(t) = P(\text{no censoring before } t)$ (or its complement $1 - C(t)$) could be done to give an impression of the proportion censored in time. Here, censoring is the “event” and a failure is a “censoring event” that prevents observation of the “event of interest.” Also, a Cox regression model with “censoring” as event can help to check whether the censoring depends on any of the covariates under consideration. If there are some variables that one may or may not include in the model (maybe they are not crucial for the question asked) then they should be included if they affect the censoring, since in this way the independent censoring assumption is relaxed to conditional independence, as discussed in Section 2.2.

An important feature of hazard models is that they can be used exactly as described in Section 2 by *formally censoring for the competing events (including all-cause death)*. This is not a violation of the independent censoring assumption, the point being, as mentioned above, that the joint likelihood function for both the event of interest and the competing events *factorizes* and the factor corresponding to the intensity for the event of interest has the same form as it would have had if competing events were regarded as censoring events.¹² In such situations one should carefully consider if the (cause-specific) hazard for the event of interest properly answers the scientific question or whether one needs to go beyond this hazard model (see Sections 4 and 5).

- *Time-dependent covariates.* When defining the model in (4), we assume that the $Z_i(t)$ are continuously measured and, thus, available at all times t , for which subject i is at risk.

A feature of the partial likelihood estimation method for the Cox model is that the values of time-dependent covariates are needed for everyone at risk at all the event times, cf. Equation (5). Some extrapolation or other ways of predicting the value of a time-dependent covariate at event times based on *past* observations ($Z(s), s \leq t$) may be needed.²⁴ In practice, most recently observed values of $Z(t)$ are typically carried forward until the next value is observed. However, such last-value-carried-forward approach can induce some bias toward the null if the current hazard depends truly on the current (unknown) covariate value.^{25,26} A more advanced approach for internal time-dependent covariates which are not measured at all times uses joint longitudinal-survival models^{27,28} to obtain estimates of β , and also allows the possibility that the observed $Z(t)$ is measured with error. Note that, for external time-dependent covariates, extrapolations are sometimes not needed since, for example, current age can be calculated based on age at baseline.

If the value of a time-dependent covariate at t captures the past history ($Z(s), s < t$) then issues of incomplete data may arise with delayed entry if ($Z(s), s < V$) is not observed. Such a problem must be discussed on a case-by-case basis.

Covariates that change shortly before the endpoint should be viewed with particular suspicion. A common example is a change in medication in the last 1 or 2 weeks before death; such changes often occur when a patient enters terminal hospice care for instance. The most serious examples of such “anticipation” involve *reverse causality bias* where a change in $Z(t)$ occurs *because of* early symptoms of the event of interest.²⁹ In some applications it may be therefore more plausible that the current hazard depends on the past rather than most recent value(s) of a time-dependent covariate implying either lagged or cumulative effects that would require more complex modeling.^{30,31}

After having fitted a Cox model one should consider:

1. *Check PH and the functional form.* Two basic assumptions of the model are that the coefficients β are time-fixed (PH assumption) and that the covariate effect is linear on the log hazard. Checking the PH assumption has developed into a large “industry” within survival analysis and giving a comprehensive review is beyond the scope of the present paper. Among the many methods proposed (some of which will be illustrated in the Examples Section 6) we mention those based on Schoenfeld and martingale residuals,^{32,33} graphical methods such as plots of the cumulative hazard,⁸ or through estimates of $\beta(t)$ in a time-varying coefficient Cox model.²⁰ For relaxing the linearity assumption, one may wish to use simple transformations like the logarithm or the square root or, alternatively, flexible modeling using, for example, splines.¹⁷ For continuous covariates, functional form (ie, nonlinear effects) should ideally be investigated jointly with assessing possible violation of the PH hypothesis (ie, their “time-varying effects”). Indeed, a failure to account for a time-varying effect may induce a “spurious evidence” of nonlinearity and vice versa.^{34,35}

Another question is what to do if model assumptions seem to be violated. Here, the answer must depend on what are the consequences of the model violation. In a classical epidemiological “exposure-confounder” situation, if the assumptions do not hold for some of the *confounders*, one may wish to perform a sensitivity analysis. Specifically, to relax the PH assumption, one can introduce time-varying effects $\beta(t)$ in the model (see (7)) or use a stratified model and if the results for the exposure in the sensitivity analysis do not change materially, the assumption may not be problematic. If, on the other hand, the assumptions do not hold for the exposure, one should carefully think about the study question and then employ extensions of the basic model if needed. In such cases modeling the time-varying hazard ratio may yield important insights into the role of a given exposure or risk factor. Note that violation of the PH assumption may be some times induced by a failure to include in the Cox model a strong predictor of the outcome.^{36,37}

2. *Reporting.* Users of the Cox model often report the regression coefficients, but not the baseline hazard. This means that measures like absolute risk cannot be retrospectively obtained from published reports. This is insufficient because the regression parameters figuring in the partial likelihood only give information about the hazard ratios and the relevance and importance of the hazard ratios at any follow-up time depends on the concurrent values of the baseline hazard.

The discrete nature of the estimated baseline hazards in the Cox model makes it hard to compare the hazards. For survival data, the estimated survival probability $\hat{S}(t|Z)$ can be used to quantify the effects of Z . This is only possible if $Z(t)$ is a time-fixed or an external covariate (see Section 4). Predicted survival curves for a population may be calculated by averaging over the observed covariate distribution (using the g-formula, see Section 5, Equation (13)).

3. *Interpretation* Three phenomena hamper the interpretation of the results (hazard ratios) of a Cox model:

- *Noncollapsibility.* It is frequently seen that the effects, β , in a Cox model gradually decay with time toward 0. This happens even if the true effect (i.e., given all relevant covariates) is perfectly constant over time if a covariate with an effect on the hazard is omitted, even if that covariate is completely independent of the other covariates. Thus, if

the correct model is $\lambda(t) = \lambda_0(t) \exp(\beta_1 Z_1 + \beta_2 Z_2)$ then a reduced model $\lambda(t) = \tilde{\lambda}_0(t) \exp(\tilde{\beta} Z_1)$ cannot hold even if Z_1 and Z_2 are independent, so that Z_2 is not considered a confounder for Z_1 .³⁸ The noncollapsibility suggests that PH can only be seen as a working hypothesis allowing a simple structure. It can be noted that the logistic regression model for a binary response variable suffers from the same problem, while the additive hazards model does not.

- *Competing risks.* If there are *no other causes of death* then the function obtained from the Cox model (or any other hazard model) using the formula $F(t|Z) = 1 - \exp(-\Lambda(t|Z))$ gives the risk of failure up to t and, in this case, hazard ratios reflect (they are in one-to-one correspondence with) ratios between absolute risks. However, if individuals may die from causes other than the one of primary interest, no such direct interpretation in terms of probability of the event is possible. Directly using the above formula will over-estimate the probability of the event of interest.⁷ Instead, one should fit the Cox models for all transitions (cause-specific hazards) and interpret the collective results through study of the absolute risk (cumulative incidence function), see Section 4. Hazard ratios from cause-specific Cox models do *no longer* necessarily reflect the absolute risk ratio.
- *Lack of causal interpretation.* Suppose the estimated hazard ratio for a treatment variable changes over time in such a way that, before some time point τ , it is less than 1 (suggesting a beneficial effect) and after τ it is equal to 1, that is:

$$\lambda(t) = \lambda_0(t) \exp(\beta_1 ZI(t < \tau) + \beta_2 ZI(t \geq \tau)),$$

with $\beta_1 < 0$, $\beta_2 = 0$. Even though this may be a correct model for the data it would be incorrect to assert that “treatment only has an impact before time τ .” This is because the hazard does not provide a “causal contrast.”^{39–41} We will also elaborate on this point in Section 5.

3.4 | Time-dependent covariates and immortal time bias

In the process of creating the dataset it is all too easy, unfortunately, to ignore the fact that a covariate is time-dependent and treat it as time-fixed. This is a common source of “immortal time bias” and may be the single most prevalent reason for invalid survival analyses in the literature.

It is crucial that only information reflecting covariate values observed before time t (ie, from the “past” at t) is used to define the value of a variable $Z(t)$ at time t . Thus, even though later information pertaining to changes that occurred *after* t may be available to the investigator at the time of analysis, only information for a given subject that reflects changes that occurred before time t can be included as part of that subject’s past at time t . We give a couple of the most common examples with invalid $Z(t)$, but these do not nearly exhaust the possibilities.

- A common example is to group patients at time $t = 0$ according to the use of drugs or treatments at any time during the follow-up even if many might have started their use only at some time after time $t = 0$ (“ever treated” vs “never treated”).

For example, in the lithium and dementia study, as mentioned above, one cannot say that only individuals that are never treated with lithium throughout the study serve as a control and thus create a time-fixed covariate “exposure.” While the information that an individual has not been treated throughout our follow-up is known at the time of data analysis, it could not be known when the patients entered the cohort. The exposure should, thus, be coded as a time-dependent covariate that starts with 0 for all individuals that were sampled from the general population, but may switch to 1 at a subject specific time t_i if subject i did start lithium treatment at that time. The alternative, that is, to treat all individuals sampled from the general population as unexposed regardless of what happens to them later will bias the comparison in the direction of “protective effect” of lithium exposure, as explained in Section 2.2 above.

Another classical example of the same problem is the Stanford heart transplant data,⁴² which included patients who were eligible for heart transplant. The event of interest was death and the focus was how the survival of transplanted subjects compares to that of not transplanted. When the data were first analyzed,⁴³ the patients were divided into two groups (“ever transplanted” vs “never transplanted”), the group membership was wrongly represented as a time-fixed covariate. As it turned out later with correct analysis, original results which suggested that the transplant is beneficial were solely due to the immortal time bias.

- A similar situation occurs when studying a trait that develops in time (eg, with time patients develop side effects or, with time, patients may respond to chemotherapy). Here, the value of their covariate starts as 0 and may become equal to 1 later. This automatically implies that individuals must survive at least some time to develop the trait, the early

deaths are hence more likely to occur in patients without the trait. Considering the value of the covariate as time-fixed and wrongly coding it as 1 already at the start (ever developed a trait or ever had side effects), will mis-attribute the portion of event-free survival time from the "unexposed" (no trait yet) to the "exposed" group and, thus, underestimate the hazard ratio associated with having the trait.

- A further example is to model the total dose received during the entire follow-up period as a time-fixed variable. Redmond et al⁴⁴ investigated a claim of Bonadonna and Valagussa⁴⁵ that disease-free survival improved with increased total amount of drug received, and found it to be entirely due to immortal time bias because the patients who died early could not have accumulated high doses. The false result is not benign, since it would encourage providers to continue full dose treatment in the face of dose-limiting toxicities, leading to increased morbidity, suffering, and possibly even death.

In all of the above cases, creation of a well-defined time-dependent covariate where $Z(t)$ does not depend on any of $Z(s)$, $N(s)$, or $Y(s)$ for any $s > t$ repairs the bias.

4 | PREDICTION USING HAZARD MODELS

Even though the intensity discussed in previous sections provides a useful framework for statistical modelling it may be hard to explain the model results to the general public. Communication is usually easier in terms of the *absolute risk*, that is, the probability of the event occurring in some interval or, more generally, the probability of being in a certain *state* by time t . For estimation of the absolute risk it now becomes crucial to consider if other (competing) events may occur, thereby preventing the event of interest from happening, see Figure 1. In general, *all* transition hazards out of the initial state, 0, are needed to estimate absolute risks.

4.1 | Prediction in the absence of competing risks

In the case of no competing risks, there is only one hazard function in the model and the absolute risk for the interval from 0 to t is obtained directly from that hazard:

$$F(t|Z) = 1 - S(t|Z) = 1 - \exp(-\Lambda(t|Z)), \quad (10)$$

provided there are no time-dependent covariates. Absolute risk is often used to describe survival or recurrence-free survival in clinical cohorts of patients treated for cancer or other life-threatening diseases.

4.1.1 | Prediction from $t = 0$ onwards

This is relevant when the time origin is well-defined. In clinical cohorts it can be the time of diagnosis or start of treatment. In population cohorts it can be a fixed value of age. The prediction is given by the survival probability $S(t|Z)$ where Z contains the relevant information available at $t = 0$.

- *Using the Cox model:* The PH assumption is often not satisfied over the whole time range. That could be fixed by introducing time-varying effects of the covariates (or by finding another model that fits the data better, for example, an additive hazards model). If the focus is on one particular value t_s (eg, the 5-year survival in cancer), a surprisingly robust estimator of $S(t_s|Z)$ can often be obtained by applying administrative censoring at t_s and using a simple Cox model with the effects β fixed in time provided t_s is not too large.⁴⁶
- *Direct modeling:* If there were no censoring before t_s , the survival probability $S(t_s|Z)$ could be directly estimated by models for the binary outcome $I(T^* > t_s)$. There is a choice of link function: probit, logit, and complementary log-log. The latter measures the effect on the same scale as the PH model. Models can be fitted using full maximum likelihood or estimating equations approaches. Censoring before t_s can be handled by modeling the censoring distribution and using inverse probability of censoring weighting (IPCW) or by using pseudo-observations based on jack-knifing.⁴⁷

4.1.2 | Dynamic prediction

Predictions made at $t = 0$ need to be updated later on for those individuals that are still alive and at risk for the events of interest. First of all, the survival probabilities have to be replaced by the conditional probability $P(T^* > t | T^* \geq t_{\text{pred}}, Z(t_{\text{pred}}))$, where t_{pred} is the time from which a new prediction is wanted. If the model for the hazard is perfect, the conditional probability can directly be computed from the hazard using $\hat{\Lambda}(t|Z(t_{\text{pred}})) - \hat{\Lambda}(t_{\text{pred}}|Z(t_{\text{pred}}))$ for $t \geq t_{\text{pred}}$. However, it may be hard to make models that are valid over the whole time range. Therefore, an alternative is to develop a new model using the data of the individuals still alive at t_{pred} . If there is a fixed prediction window t_s , the conditional survival $P(T^* > t_{\text{pred}} + t_s | T^* > t_{\text{pred}}, Z(t_{\text{pred}}))$ can be estimated robustly by the methods discussed above. Prediction later on is known as *dynamic prediction* and the approach of building new models using the individuals at risk at t_{pred} is known as *landmarking*.⁴⁸ This is also of interest more generally when there is *delayed entry*, that is, individuals entering the cohort at $V > 0$. In this case, the hazard can be hard to estimate around $t = 0$, since only few individuals may be at risk early on. Hence, predictions are hard to make at $t = 0$, but conditional survival probabilities could be estimated reliably later in the follow-up. That might be particularly relevant for analyses that use age as the time axis.

4.1.3 | Prediction exploiting time-dependent covariates

Dynamic predictions using landmarking can thus be used when the PH assumption does not give a reasonable description over the entire time range. The technique may, however, be even more useful when doing predictions based on a model with time-dependent covariates. A hazard model with time-dependent covariates $Z(t)$ for which the trajectories are still unknown at $t = 0$ can be useful when the aim of modeling is to describe the processes behind the hazard, but it is no longer simple to calculate the survival probabilities from the hazard using the relationship $S(t) = \exp(-\Lambda(t))$. This means that such a model cannot be used for predictions at $t = 0$. However, they can still be useful because the history of $Z(t)$ before t_{pred} can be informative for the future of the process. Therefore, such predictions can be based on landmark models. The history of $Z(t)$ up to t_{pred} is summarized in a single statistic that is used as time-fixed covariate in the prediction model. The simplest approach is to use the last observation before t_{pred} . This approach does not satisfy the consistency condition that a prediction model at two different times should be compatible,^{49,50} for this reason the models used in landmarking should be regarded as working models. However, the approach can be extended to give better predictions if more flexible prediction models from the landmark time are used, and more than just the last observation of $Z(t_{\text{pred}})$ is used to represent the effect of $Z(t)$, including for example, cumulative effects.⁵¹

Another way is to develop a joint model for $Z(t)$ and $\lambda(t|Z(t))$ and estimate survival probabilities by conditioning on the history of $Z(t)$ at t_{pred} and $T^* \geq t_{\text{pred}}$ in the joint model. Estimation for such models can be challenging⁵² and while such an approach has a better theoretical justification and is efficient, there can be concerns about the robustness.

4.2 | Prediction with competing risks

Very often intensity models are used for an event that does not include all-cause mortality. This was for example the case in the lithium and dementia study. However, in the presence of competing risks, naively inserting the estimated hazard into Equation (10) will produce an upwards biased estimate of the absolute risk (cumulative incidence). This is because, by treating competing events as censorings, one pretends that the target population is one where the competing events are not operating and therefore neglects the fact that subjects who have died from competing causes can no longer experience the event of interest. In such a situation it is necessary to estimate the *absolute risk*. This is the probability of experiencing event 1 at a certain time point and it depends not only on the hazard of this event but also on the probability of having survived up to this time point. And as this survival probability depends on the hazards of all events, this implies it is necessary also to estimate the intensity of the competing events. If the cumulative hazard for the competing events is $\Lambda_{02}(t|Z)$ then the absolute risk of event 1 is given by

$$F_1(t|Z) = \int_0^t S(u|Z) d\Lambda_{01}(u|Z) = \int_0^t \exp(-\Lambda_{02}(u|Z) - \Lambda_{01}(u|Z)) d\Lambda_{01}(u|Z). \quad (11)$$

Without covariates, the estimator obtained by plugging-in Nelson-Aalen estimates for the cumulative hazards in (11) is known as the *Aalen-Johansen estimator*.⁸ The intuition behind the middle term of (11) is that the integrand, the survival

probability, $S(u|Z)$ of surviving to time u , times the conditional probability, $d\Lambda_{01}(u|Z)$ of dying from cause 1 at time u given survival till u , is the (unconditional) probability of dying from cause 1 at time u ; $F_1(t|Z)$, the probability of dying from cause 1 by time t is then obtained as the sum (integral).

It is also possible to set up direct regression models for $F(t|z)$, for example, using the Fine and Gray regression model⁵³ but a further discussion of such methods is beyond the scope of the present paper. We will, however, exemplify the use of the absolute risk in the examples in Section 6.

5 | ISSUES IN CAUSAL INFERENCE

“Causality” may be defined in a number of different ways but the most commonly used approach is based on potential outcomes and randomized experiments.⁵⁴⁻⁵⁶ This is because a well-conducted randomized experiment allows a causal interpretation of the estimated treatment effect. However, also in certain observational studies a causal interpretation of the effect of a nonrandomized exposure is of interest. Two classic examples where randomization cannot be employed are (i) post-marketing studies of potential “adverse effects” of medications/treatments already approved (based on earlier randomized trials that focused on their effectiveness), and (ii) environmental or occupational exposures (randomization often impossible and/or unethical). However, any attempt of causal interpretation in an observational study, obviously, requires strong assumptions. It is not the intention of this paper to go into details concerning causal inference but in the current section we will briefly discuss the topic.

First of all, causal questions are most natural and relevant for *modifiable variables* for which a hypothetical randomized study could, in principle, be done. They are less relevant for variables that you cannot change (such as sex or race). Causal parameters are typically defined as contrasts between average outcomes for the same population under the hypothetical scenarios of every one being “treated” vs every one being “untreated.” Thus, the causal risk difference at time t is:

$$\Delta(t) = P(T^*(0) \leq t) - P(T^*(1) \leq t), \quad (12)$$

where $T^*(0)$, $T^*(1)$ are the (possibly counterfactual) survival times “under no treatment,” vs “under treatment.” The causal parameter, $\Delta(t)$ in Equation (12) is directly estimable in a randomized study and may be estimable based on observational data under a set of assumptions, including “no unmeasured confounders”—a condition that can, obviously, never be tested based on the available data.

Under these assumptions, one way of getting from a hazard model to an estimate of the counterfactual risk, had all subjects in the population been treated with treatment $a = 0, 1$, is to use *inverse probability of treatment weights*; another is to use the “g-formula.”⁵⁶ The latter works, as follows. If the hazard model for given treatment A and confounders Z leads to an estimated absolute risk of $\hat{F}(t|A, Z)$ then the estimate of $P(T^*(a) \leq t)$ using the g-formula is

$$\hat{P}(T^*(a) \leq t) = \frac{1}{n} \sum_i \hat{F}(t|a, Z_i), \quad a = 0, 1. \quad (13)$$

That is, the risk is predicted for each given subject under treatment $A = a$ given his or her observed covariates and then *averaged* over the sample $i = 1, \dots, n$. Formula (13) is applied separately for each treatment ($a = 0$ vs $a = 1$) and the estimate of $\Delta(t)$ is obtained.⁵⁷ Note that the g-formula is useful for predicting average risk in a sample even though a causal interpretation is not aimed at.

Another use of the g-formula can yield an estimate of the number of events “attributable to” a certain modifiable risk factor, A . This number is given by the difference between the “total risk” *observed* in the population before time t : $\sum_i \hat{F}(t|A_i, Z_i)$, and that expected *if every one was unexposed* ($A_i = 0$): $\sum_i \hat{F}(t|0, Z_i)$. When doing this for a number of risk factors, these may be ranked in a way that also accounts for their prevalence in the population.

Though a hazard model may, thus, be useful both for describing associations between covariates and a time-to-event outcome and serving as a useful step toward estimating a causal contrast like (12), it does not itself provide a causal contrast. This may be seen, as follows. Recall the intuitive definition (1) of the hazard function for survival data:

$$\lambda(t) = P(T^* \leq t + dt | T^* > t) / dt.$$

This shows that contrasts based on the hazard functions for the counterfactual outcomes $T^*(0)$, $T^*(1)$, for example, the hazard ratio at time t ,

$$\frac{\lambda^1(t)}{\lambda^0(t)} = \frac{P(T^*(1) \leq t + dt | T^*(1) > t)}{P(T^*(0) \leq t + dt | T^*(0) > t)},$$

are not directly causally interpretable since they contrast different subpopulations: those who survive past t under treatment ($T^*(1) > t$) and those who survive past time t under no treatment ($T^*(0) > t$). For this reason, a statement saying that “treatment only works until time τ but not beyond” in a situation with $\beta(t) < 0$ for $t < \tau$ and $\beta(t) = 0$ for $t > \tau$ is not justified.^{39–41} A similar remark holds for hazard differences—see Martinussen et al⁴¹ for further discussion of this point and other issues with causal interpretations of hazard contrasts.

A special problem with causal inference for survival data is *time-dependent confounding/mediation* where a time-dependent covariate both affects future treatment and survival outcome and is affected by past treatment. For this situation, special techniques are needed to draw causal conclusions concerning the treatment effect.^{56,58}

6 | ILLUSTRATIVE APPLICATIONS

In this section, we illustrate the points given in the paper with three real data examples. A more detailed analysis (along with code in R statistical software) is provided in Appendix S1.

6.1 | Peripheral arterial disease

PAD is a common circulatory problem in which narrowed arteries reduce blood flow to peripheral limbs, often the legs. It is also likely to be a sign of a more widespread atherosclerosis, and subjects manifesting the disease carry an increased risk for atherothrombotic events. The PAD dataset contains the results of a Slovene study reported in Blinc et al.^{59,60} Briefly, the study was conducted by 74 primary care physicians-researchers (GPs), who recruited subjects with PAD along with age and sex matched “controls” without PAD. Yearly examination visits were planned with a total of 5 years of follow-up. The final study included 742 PAD patients and 713 controls, with baseline data for each subject, measurements at each visit, and endpoints. Important endpoints are death, either due to cardiovascular disease (CVD) or other causes, nonfatal CVD endpoints of infarction and stroke, and patient interventions attributed to the disease such as revascularization procedures. All the individuals in the study were treated according to the latest treatment guidelines and the goal of the study was to study survival of the patients with PAD (in comparison to controls) despite optimal treatment.

Endpoints: Most of this analysis will focus on death as the outcome of interest. The causes of death are split into two groups: CV or other (non-CV) and, in addition, we will also consider all CV events (stroke, infarction or death) as an outcome for modeling.

Inclusion criteria, follow-up and censoring: With most GPs, the follow-up visits of the patients followed a yearly plan, though, in practice, the visits tended to be moderately delayed, with time to the fifth visit ranging from 4.8 to 6.8 years. Most data on patients were recorded at the time of their visit, with the exception of deaths which were reported as they occurred (along with all other events that occurred since the last visit).

Because of between-physician differences in whether patients were followed after 5 years, to avoid possibly nonindependent censoring, all individuals alive at 5 years after enrollment were censored at that time.

Time axis, basic survival analysis: For the PAD patients the time since diagnosis is a natural time axis as it represents both the progression of disease and treatments for the disease. Survival curves for the control subjects serve as a comparison outcome of similarly aged subjects without the disease, but do not have a natural stand-alone interpretation. Figure 2 contains the overall Kaplan-Meier curves for PAD and control subjects, male and female (deaths of any cause are considered as the outcome). The survival is higher for females than for males, which is no surprise given a mean age at entry of 65 years, and is lower for PAD subjects than for the age- and sex-matched controls. The right hand panel shows the curves on age as the time axis, a very similar pattern can be observed. When using age, the left hand portion of a survival curve can often be highly variable due to the small number of the patients at risk at a young age and this early high variability can then affect the entire curve. To avoid this, we estimate conditional survival $P(T^* > t | T^* > t_0)$ for $t > t_0$, with t_0 chosen so that the risk set is large enough and most of the information of interest is included. In the case of PAD, we choose $t_0 = 55$ years.

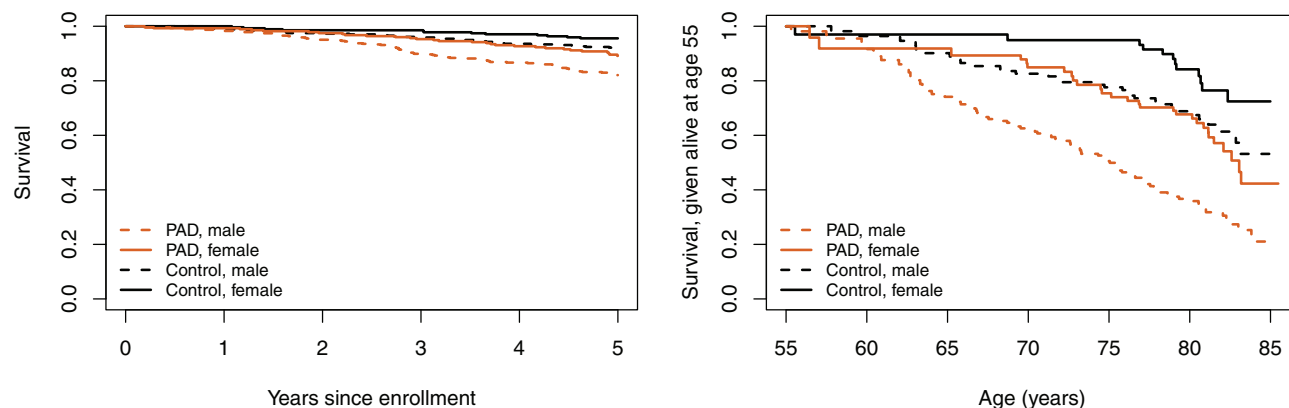


FIGURE 2 Kaplan-Meier estimates of cumulative probability of survival with respect to time since enrollment (left) and age (right) [Colour figure can be viewed at wileyonlinelibrary.com]

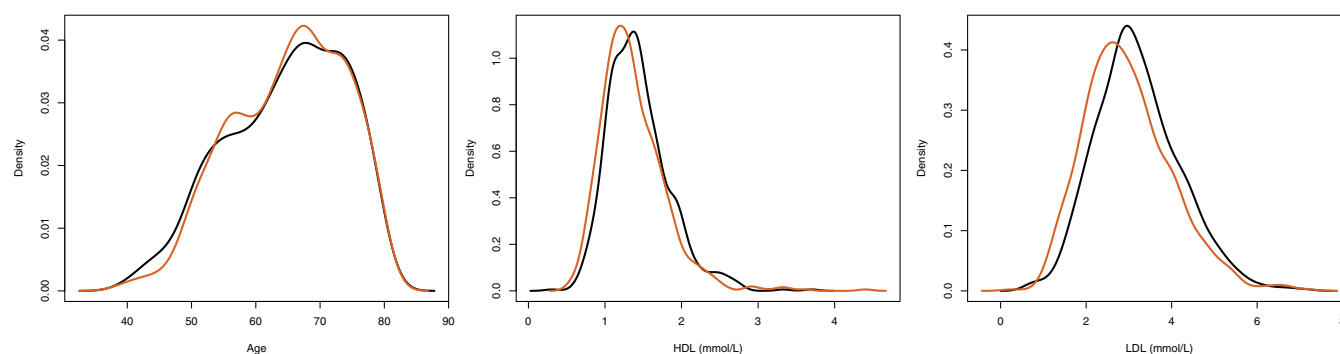


FIGURE 3 Distributions of continuous variables with respect to peripheral arterial disease (PAD) (red=PAD, black=Control) [Colour figure can be viewed at wileyonlinelibrary.com]

Hazard regression models

We next study how the covariates affect the hazard of dying.

Covariates: We will be interested in PAD, sex (38% women), age, and later also in Low-density lipoprotein (LDL) and High-density lipoprotein (HDL). The distribution of the continuous covariates with respect to PAD is given in Figure 3. By study design there should be no difference in the age distribution, HDL and LDL are slightly lower for the PAD subjects. When used as a covariate, age will be expressed in decades to give a coefficient of a more interpretable size.

In the analysis presented in Table 1, we focus on the effect of PAD and the effect of sex in each PAD subgroup. First, we fit a Cox model with time since enrollment as the time axis. Knowing that age is a strong predictor, we include it in the model (ie, age/10 is used as a covariate). We learn that patients with PAD have a 2.4 times higher hazard than the controls, and that male sex and 10 additional years of age each increase the hazard by approximately 2-fold. The effect of both sex and age is very similar in both groups (patients and controls).

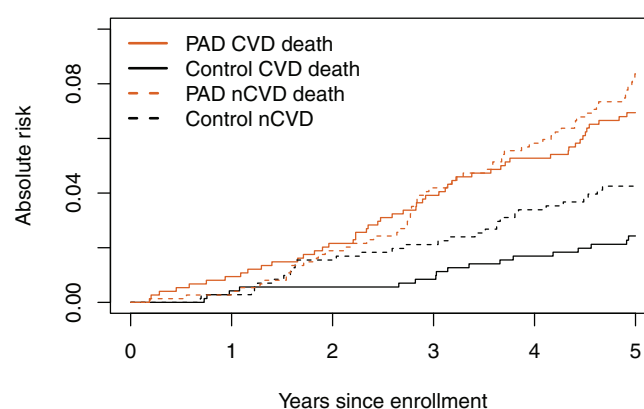
Alternatively we can use age as the time axis and add time since enrollment (FU time) as a possible predictor. Using age as the time axis the effect of male sex is a 2-fold hazard increase, just as before. The time-dependent variable years-since-enrollment compares the hazard of death for subjects with more study years to those recently enrolled, and shows an increase in death rates over time (HR=1.2 per year).

By choosing one time axis and adding the other into the model as a covariate, the interpretation of the HR for sex becomes equal (the HR for two patients of the same age and same time since enrollment) under the condition that the assumptions of the linearity (and PH) of the covariate are met. To avoid the problem of choosing the time axis and adding assumptions, the Poisson model that allows multiple time axes can be used instead. To this end, we assume the baseline hazard constant within yearly intervals of time since enrollment and five-year intervals of age. The results are given in the last two rows of Table 1. We can see that, in our case, all three approaches coincide well, so, the possible violations of the assumptions of the different options had no effect.

TABLE 1 Estimated hazard ratios (HR) and 95% confidence intervals (CI) in models with different time axes, fitted with Cox or Poisson model. The last column reports the *P*-value for interaction of each covariate with group (PAD or control)

	Overall		PAD		Control		P
	HR	95% CI	HR	95% CI	HR	95% CI	PAD vs C
Time since enrollment axis, Cox model							
PAD	2.40	(1.71, 3.37)					
Sex (m vs f)	2.00	(1.40, 2.86)	2.01	(1.31, 3.08)	1.97	(1.02, 3.79)	0.96
Age (per 10 years)	1.93	(1.57, 2.37)	1.91	(1.49, 2.45)	1.98	(1.36, 2.89)	0.88
Age axis, Cox model							
PAD	2.40	(1.70, 3.37)					
Sex (m vs f)	2.02	(1.42, 2.90)	2.01	(1.31, 3.08)	2.01	(1.04, 3.88)	1.00
FU (per 1 year)	1.18	(1.05, 1.33)	1.20	(1.05, 1.38)	1.12	(0.91, 1.39)	0.61
Both time axes, Poisson model							
PAD	2.38	(1.70, 3.35)					
Sex (m vs f)	2.01	(1.41, 2.88)	2.03	(1.33, 3.11)	1.97	(1.02, 3.81)	0.95

Abbreviations: PAD, peripheral arterial disease. FU, time since enrollment.

FIGURE 4 The probability of dying due to cardiovascular (solid line) or other reasons (dashed line) with respect to peripheral arterial disease (red=patients, black=controls) [Colour figure can be viewed at wileyonlinelibrary.com]

Competing risks and time-dependent covariates

The analysis so far considered all causes of death equally, but the CV deaths are of particular interest. In 5 years, 159 patients died, 68 of these due to CV reasons. Figure 4 presents the Aalen-Johansen estimator of the absolute risk (also known as the cumulative incidence function). The estimated 5-year survival probability of PAD patients is 84.5 % and we can see that 6.9% of the PAD patients are estimated to have died due to CV reasons and 8.4 % due to other reasons. Both the probability of CV but also that of non-CV death are considerably greater than in the control group.

We wish to explore the effect of our basic covariates (sex, age, and PAD) and cholesterol (LDL and HDL) on the hazard for CV death or major CV events. Hazard models for a particular endpoint can be fitted by censoring all “other cause” deaths. The results for Cox models on the time-since-enrollment axis are given in Table 2.

- A: As before, we see that male sex and higher age increase the hazard, we also see that PAD is a strong risk factor, the hazard of PAD patients is almost three times higher than that of the controls. Neither LDL nor HDL values at baseline seem to have an important effect.
- B: We now include all the information available - the values of HDL and LDL were updated on a yearly basis (if missing, the last value was carried forward), that is, we use them as time-dependent variables. We can now observe a much larger effect of HDL, patients whose HDL is lower by 1 mmol/L have a 4.7 ($\approx 1/0.21$) times higher hazard. We can also observe a more pronounced effect of LDL. Its direction may seem counterintuitive, but may be due to the fact that

TABLE 2 Estimated hazard ratios (HR) and 95 % confidence intervals (CI) in Cox models. A: CV deaths, baseline HDL and LDL; B: CV deaths, time-dependent HDL and LDL; C: all CV events, time-dependent HDL and LDL

	A		B		C	
	CV death		CV death		CV events	
	Time-fixed		Time-dependent		Time-dependent	
	HR	95%CI	HR	95%CI	HR	95%CI
Peripheral arterial disease	2.87	(1.65-5)	2.40	(1.37-4.20)	2.27	(1.57-3.28)
Sex (m vs f)	1.67	(0.97-2.88)	1.36	(0.79-2.35)	1.90	(1.28-2.81)
Age (per 10 years)	1.93	(1.40-2.66)	2.01	(1.45-2.77)	1.54	(1.25-1.90)
HDL (mmol/L)	0.74	(0.39-1.41)	0.21	(0.10-0.48)	0.48	(0.29-0.79)
LDL (mmol/L)	0.92	(0.72-1.18)	0.76	(0.57-1.01)	0.88	(0.73-1.07)

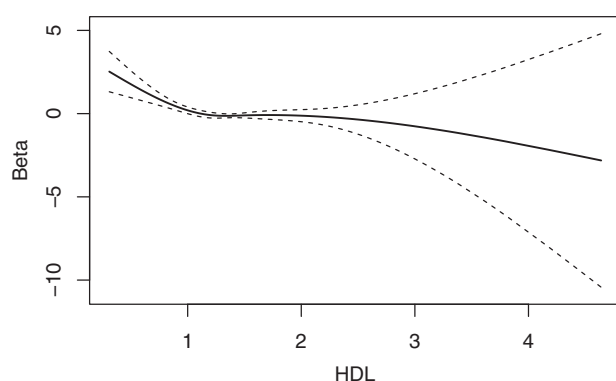


FIGURE 5 The effect of HDL on the hazard of having a cardiovascular event modeled by using restricted cubic splines in a multiple Cox regression model (an extension of model C)

patients at a higher risk have lower target values of LDL and hence the lower LDL may be a proxy for the higher risk patients.

- C: This model regards not only CV death but also stroke and infarction as events. The effects of the covariates do not change much, but all the SEs have decreased as the number of events increased to 142.

To check whether the above interpretation makes sense, we further examine the goodness-of-fit of the models.

We focus on model C, which uses all the information available. Adding a spline to the model, that is, replacing β HDL by $s(\text{HDL})$, we can see that the linearity of HDL may be problematic. The protective effect increases with the value of HDL but may level off for values above approx 1.5 mmol/L, see Figure 5 (the huge confidence interval beyond 2 mmol/L is due to very few individuals with HDL above 2). Allowing HDL to be nonlinear, the Schoenfeld's residuals test for PH indicates no further issues.

The calculation of the absolute risk is based on model A, as only the baseline information can be used for prediction. Unlike in the case of pure hazard modeling, the other causes of death cannot be simply censored—to estimate the probability of dying due to cancer, the hazard of dying due to other causes must be estimated as well, see Table 3. The absolute risks of two individuals, one aged 58 and the other 72 (25th and 75th percentile of age) and median values of lipids are plotted in Figure 6.

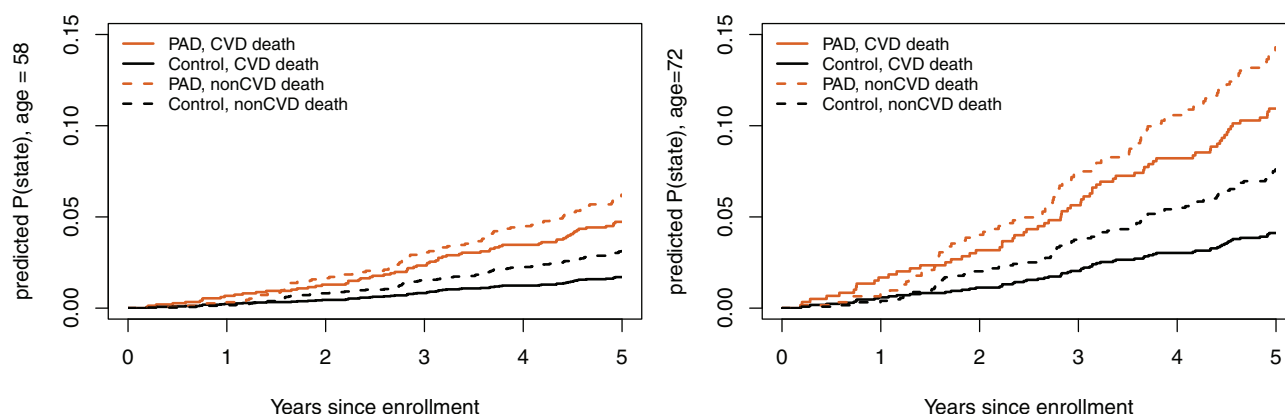
To conclude, we have seen that the PAD remains a strong risk factor despite following the latest treatment guidelines. This is true regardless of whether we focus on all major CV events or on CV death only. Old age and too low HDL are further associated with a higher event rate.

6.2 | Nonalcoholic fatty liver disease

NAFLD is defined by three criteria: presence of greater than 5% fat in the liver (steatosis), absence of other indications for the steatosis (such as excessive alcohol consumption or certain medications), and absence of other liver disease.⁶¹ NAFLD

TABLE 3 Estimated hazard ratios (HR) and 95% confidence intervals (CI) for other cause mortality. Baseline HDL and LDL

	Time-fixed, other cause	
	HR	95% CI
Peripheral arterial disease	2.04	(1.31–3.19)
Sex (m vs f)	2.12	(1.29–3.50)
Age (per 10 years)	1.93	(1.45–2.56)
HDL	0.82	(0.43–1.55)
LDL	1.02	(0.83–1.26)

**FIGURE 6** The absolute risk of dying of cardiovascular (solid line) and other causes (dashed line) by peripheral arterial disease status (patients=red, controls=black) and age (left graph 58, right graph 72). The values of baseline HDL and LDL are 1.3 and 3, respectively [Colour figure can be viewed at wileyonlinelibrary.com]

is currently believed to be responsible for almost one-third of liver transplants and its impact is growing. It is expected to be a major driver of hepatology practice in the coming decades.⁶² The study included all patients with a NAFLD diagnosis in Olmsted County, Minnesota between 1997 and 2014 along with up to four age- and sex-matched controls for each case.⁶³ (Note that some changes to the public data have been made to protect patient confidentiality; analysis results here will not exactly match the original paper).

The goal of the study is to investigate whether NAFLD subjects are at increased mortality risk compared to the general population, and if so the amount of increase. Only a minority of subjects are tested for NAFLD since this requires an abdominal scan, and we can, therefore, only address the progression of *detected* NAFLD.

Entry time, inclusion criteria: In the PAD study, the data were collected prospectively and hence the inclusion criteria were naturally evaluated at the time of inclusion. On the contrary, the NAFLD data were collected retrospectively using existing databases and are hence more prone to mistakes regarding the time when the inclusion criteria are known.

Subjects enter the study at the age of NAFLD diagnosis or selection as a control, whichever comes first. Because NAFLD is often a disease of exclusion, a NAFLD diagnosis followed shortly by the diagnosis of another liver disease is considered a false positive. The data set is restricted to “confirmed NAFLD,” that is, if someone were diagnosed on 20 June 2001, the index date for confirmed NAFLD would be 20 June 2002, assuming that another liver diagnosis, death, or incomplete follow-up did not intervene. The follow-up of the matched control subjects also commences on the “confirmed NAFLD” date. This is important. If the matched subjects’ follow-up were started on 20 June 2001 then the control has the opportunity to die during that first year while the case does not, leading to immortal time bias.

When selecting the controls for any given NAFLD case at age a , it is very important only to use information that was available at age a for the controls. We cannot exclude subjects who have too short a follow-up (die or censored before age $a + 2$ say), will later have diabetes, or, most particularly, those who will later become NAFLD patients. Each of these is a variant of immortal time bias. In this dataset, 331 of the subjects selected as controls were diagnosed with NAFLD at a later age. Care must be taken at the time of analysis to correctly deal with these patients. The preliminary checks and figures will treat each subject’s value at study entry as fixed, the hazard models will treat it as a time-dependent covariate.

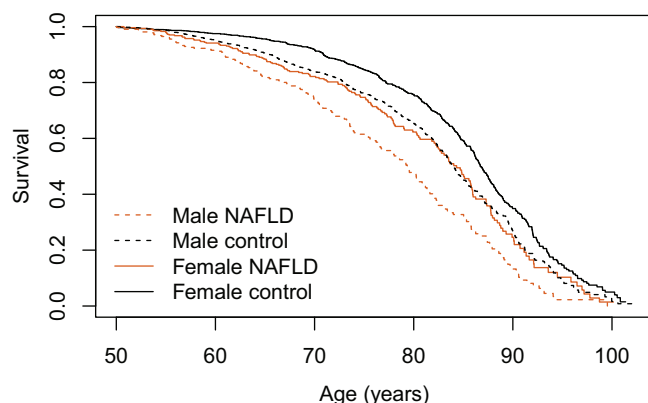


FIGURE 7 Survival curves from age 50 forward, comparing nonalcoholic fatty liver disease (NAFLD) to non-NAFLD at study entry, stratified by male/female [Colour figure can be viewed at wileyonlinelibrary.com]

	Female control	Female NAFLD	Male control	Male NAFLD
40-50	1.3	2.4	2.2	2.5
50-60	2.5	5.9	5.2	8.3
60-70	5.4	14.8	11.6	22.8
70-80	18.0	28.1	23.4	37.2
80-90	68.1	76.3	79.6	108.4

TABLE 4 Death rates per 1000 person years, split by age group, sex, and time-dependent nonalcoholic fatty liver disease (NAFLD) status

Endpoints and censoring: The primary focus of this analysis is death, which means it is relatively straightforward to handle. All the subjects in the study are administratively censored at the end of 2017, when the dataset was created. A small number has been censored due to migration, about 1% per year over the age of 50.⁶⁴

Because the publicly available NAFLD dataset does not contain dates, a plot of the censoring distribution is not particularly informative: we do not know what the result *should* look like.

Since the follow-up is as long as 20 years for some subjects, care must be taken with the independent censoring assumption - the later included subjects have a systematically lower death rate, for example, due to improved general population medical care, and are, due to the later inclusion date, followed-up for a shorter time period.

Time axis, basic survival analysis: The NAFLD is a not an acute condition, it may well exist for many years before detection. Furthermore, the age range in the study is very wide, death as the primary endpoint is highly related to age and cases and controls match with each other on age (with "time since NAFLD" not well defined for the controls). All these reasons make age the natural time axis for the NAFLD study. This approach mimics an idealized (but impractical) study which included the entire population from birth forward, with time-dependent NAFLD as the covariate.

As a first description of data, we plot the estimated survival curves for the patients by sex and NAFLD group, see Figure 7. For the latter we use the subject's NAFLD status at enrollment as a time-fixed variable. This approach is similar in spirit to using intent-to-treat in a clinical trial, in that it gives a reliable estimate but one that may underestimate the true clinical effect of a covariate. As with the PAD study we estimate conditional survival.

An alternative summary is to report the cumulative hazard (using the Nelson-Aalen estimator) by sex and time-dependent NAFLD status, see Figure 8. If the hazard is constant on the interval, the increase of the cumulative hazard on each interval is close to the death rates (proportion of deaths per person-year) given in Table 4. The hazard ratio (difference on log scale) between male patients and controls is nearly constant in time, which suggests a PH model may fit well. On the other hand, in women the NAFLD/control hazard ratio is highest at the youngest ages and decreases with age.

Hazard models: In the hazard models we can incorporate NAFLD as a time-dependent covariate. Subjects who were NAFLD at enrollment have a value of 1, controls start with a value of 0 at enrollment, some of whom turn to 1 when they are diagnosed with NAFLD at a later age. Also, the overall model is not adversely affected by the small sample issue at younger ages, so there is no need to use a restricted age range. Because NAFLD is strongly associated with obesity, we also fit models that adjust for other conditions associated with obesity: diabetes, hypertension and dyslipidemia. Fits were done overall and for males and females separately.

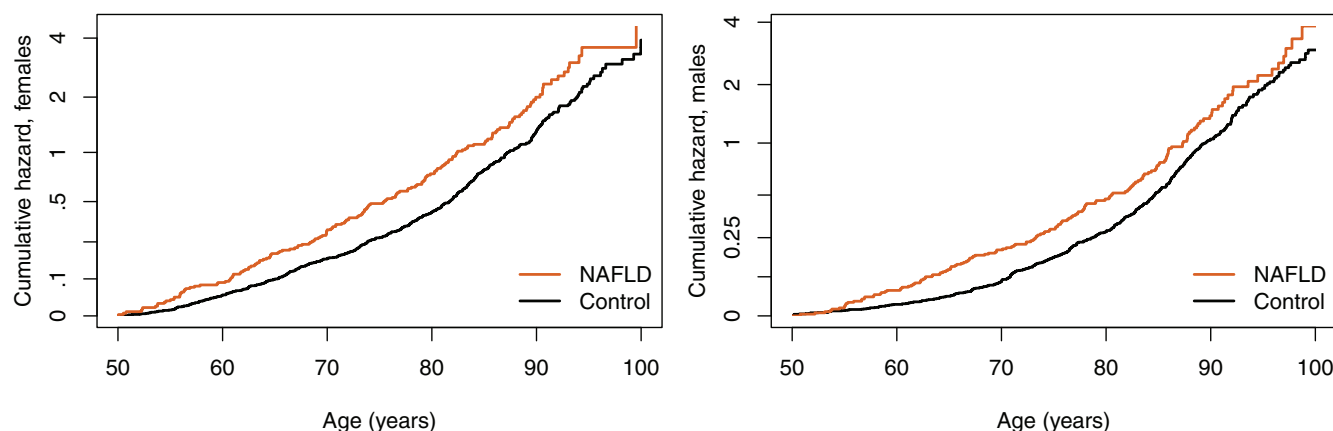


FIGURE 8 Nelson-Aalen estimates for the cumulative hazard from age 50, stratified by gender (left: females, right: males) and nonalcoholic fatty liver disease (NB: log-scale on the vertical axis) [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/sim.8757)]

TABLE 5 Estimated hazard ratios (HR) and 95 % confidence intervals (CI) from Cox models that have only nonalcoholic fatty liver disease (NAFLD) as a covariate, and models with NAFLD and covariates. The overall model is fit to all subjects with sex as a stratification variable

	Overall		Females		Males	
	HR	95%CI	HR	95%CI	HR	95%CI
NAFLD only	1.62	(1.44-1.82)	1.65	(1.39-1.95)	1.60	(1.35-1.88)
NAFLD	1.43	(1.26-1.62)	1.39	(1.17-1.67)	1.45	(1.22-1.73)
Diabetes	1.77	(1.57-2.01)	1.94	(1.62-2.32)	1.64	(1.38-1.94)
Hypertension	1.24	(1.08-1.42)	1.33	(1.10-1.63)	1.16	(0.96-1.41)
Dyslipidemia	0.68	(0.60-0.78)	0.65	(0.54-0.79)	0.72	(0.60-0.88)

Table 5 contains the estimated hazard ratios from Cox models that include all subjects, only males or only females, and for models that include only NAFLD as a (time-dependent) covariate as well as adjusting for confounders. The estimated effect of NAFLD is attenuated when adjusting for the three covariates. The higher prevalence of diabetes and other conditions explains a portion of the NAFLD effect. The overall NAFLD effect does not differ markedly for males and females.

Model checks: Since all of the covariates in the models are binary, there is no need to explore functional form. An overall test for PH based on Schoenfeld's residuals has results that mimic what was seen in the hazard plot of Figure 8: males fit the PH model well ($P = .4$) while females have significant nonproportionality ($P < 0.001$). It is interesting that the overall "average" effects, over age, are nearly the same for male and females, however. Checks of the multiple Cox model show that nonproportionality is more severe with respect to diabetes (for both males and females) and for hypertension for women, see Figure 9. The relative effect of comorbidities on death rates is higher at younger ages, but may get high again with very old age.

In conclusion, NAFLD is associated with an increased mortality compared to disease-free, age and sex matched controls. Part of this increase may be explained by the different diabetes, hypertension and dyslipidemia distributions in the two groups.

6.3 | Advanced ovarian cancer

As the last example, we consider the advanced ovarian cancer dataset. It contains follow-up on 358 subjects who were enrolled in two trials of chemotherapy for advanced ovarian cancer, conducted around 1980 by a multi-institution research network in the Netherlands. The eligibility criteria for enrollment included pathologic confirmation of advanced disease, age less than 70 years, lack of serious cardiac or renal disease, and favorable haematological status. Patients could not have a second tumor, brain metastases, or prior radiation or chemotherapy. The treatment is not of our main interest here, hence we treat this data set primarily as an observational study. The data were extensively analyzed in chapter 6 of Van

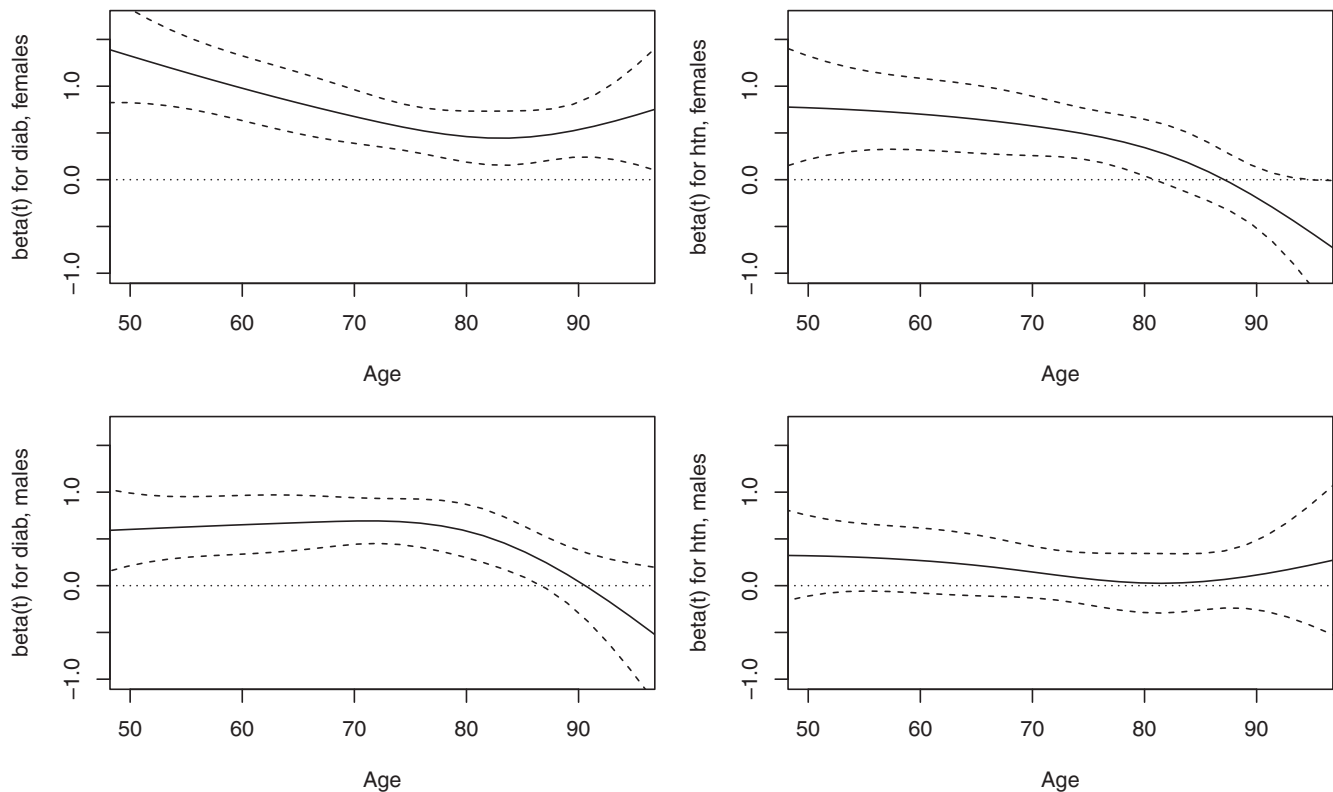


FIGURE 9 The changing effect of diabetes (diab) and hypertension (htn) in the multiple regression models. Top row: females, bottom row: males

Houwelingen and Putter,⁶⁵ and further references can be found there as well. Patient follow-up in the dataset continued for 6 years. The goal of the analysis was to predict the survival probability of patients using covariates that were recorded at baseline.

Focusing on a fatal condition such as advanced cancer in a data set that comes from a clinical trial with excellent follow-up, the basic aspects of these data are particularly simple: the sole event of interest is death of any cause, the inclusion criteria are clear and the most natural time axis is time from diagnosis as this is the time frame of most direct interest to both the patient and the care provider. The left panel of Figure 10 shows the censoring pattern for the study, which follows the expected “hockey stick” shape for a formal trial with 3 years of enrollment, 4 years of follow-up after enrollment of the final subject, and no subjects lost to follow-up. The graph shows no censoring before 4 years followed by an upward line corresponding to uniform accrual each year. The Kaplan-Meier curves give the overall pattern of survival for this cohort, see the right hand graph of Figure 10.

Regression: In our analysis, we focus on three covariates:

- FIGO: This is a staging system for ovarian cancer. Advanced ovarian cancer comprises the stages III ($n = 262$, used as reference group) and IV ($n = 96$). Stage IV patients are known to have a very poor prognosis.
- The diameter of the residual tumor after surgery, categorized as micro, <1 cm, 1–2 cm, 2–5 cm, and >5 cm, with the last category being the most frequent one ($n = 145$). We will use the ‘micro’ category ($n = 29$) as the reference group.
- Karnofsky index. A measure of the patient’s functional status at the time of diagnosis. The maximum score 10 is an indication of no physical limitations. We will regard the covariate as quantitative in the model.

The coefficients in the fitted PH model are given in Table 6.

Checking the assumptions of the model: To check the PH assumptions, we consider both the method using cumulative Schoenfeld residuals of Lin et al³³ and the smoothed Schoenfeld residuals method proposed by Therneau and Grambsch.³² Both methods agree that the PH assumption seems to be problematic for the Karnofsky score (Figure 11). The test based on cumulative Schoenfeld residuals returns a P -value of .009. The left hand graph of Figure 11 (smoothed residuals) shows

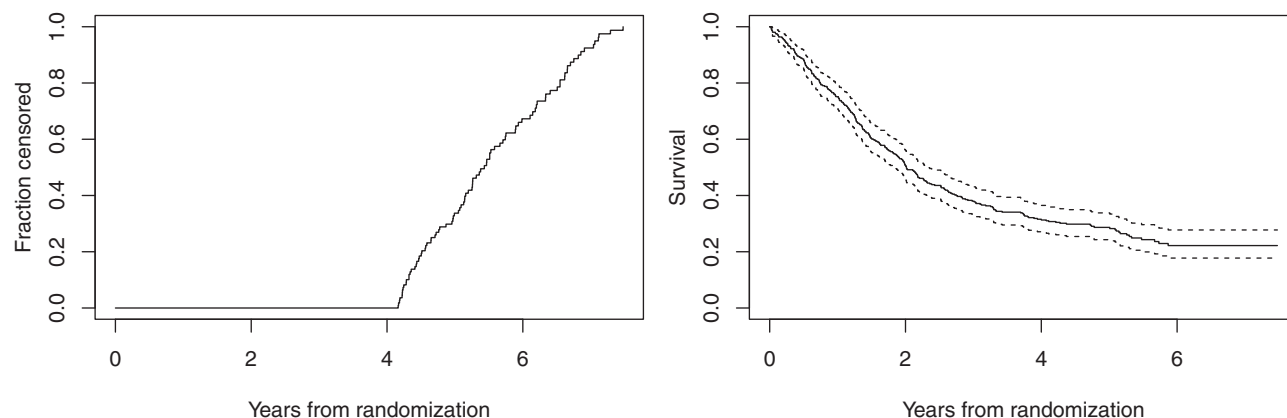


FIGURE 10 Censoring fraction and survival curve (with 95% confidence interval) for the ovarian cancer study

TABLE 6 Estimated hazard ratios (HR) and 95 % confidence intervals (CI) in the Cox model for the ovarian cancer data. The reference group for the covariate Diameter is “micro.”

	HR	95% CI
Diameter<1cm	1.38	(0.73-2.57)
Diameter 1-2cm	2.24	(1.19-4.21)
Diameter 2-5cm	2.38	(1.29-4.39)
Diameter>5cm	2.53	(1.40-4.57)
FIGO (stage IV vs. III)	1.73	(1.33-2.25)
Karnofsky index (per 1 point)	0.84	(0.75-0.93)

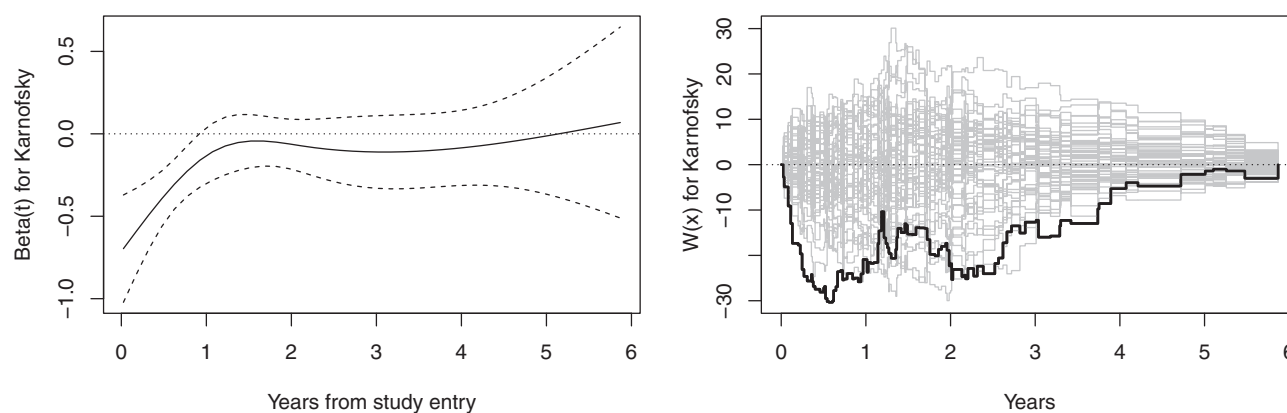


FIGURE 11 Proportional hazards plots for Karnofsky score (left: smoothed Schoenfeld residuals, right: cumulative Lin et al)

a rapid early drop in importance of the Karnofsky score, implying that baseline Karnofsky score, measured at diagnosis, is not predictive of mortality beyond the first year of follow-up, something that could be expected for advanced cancer.

Dealing with the lack of PH: One approach to deal with the violation of PH assumption is to fit a set of landmark models. In this case we consider 2-year windows and set the landmark points at 0, 1, and 2 years. By fitting a separate model at each landmark point, we allow the coefficient to change in time but also to use the most recent covariate values for the prediction (in this case there are no time-dependent covariates, so all fits use the same values). As can be seen from Table 7, the effect of the residual tumor diameter increases with size at all the landmark points, though quite some variability can be observed there (the subgroups are rather small and hence the standard errors quite large). The coefficient for the stage at baseline (FIGO) remains rather constant in time, whereas the effect of Karnofsky index quickly approaches zero (hazard ratio approaches 1), for predictions from 1 or 2 years onward, the baseline value of Karnofsky score carries no important information.

	From time 0		From 1 year		From 2 years	
	HR	95% CI	HR	95% CI	HR	95% CI
Diameter<1 cm	1.31	(0.5-3.3)	1.63	(0.7-4.1)	2.73	(0.8-9.4)
Diameter 1-2 cm	2.92	(1.2-7.1)	2.89	(1.2-7.2)	2.17	(0.5-8.7)
Diameter 2-5 cm	3.04	(1.3-7.2)	2.75	(1.1-6.7)	3.55	(1.0-12.7)
Diameter>5 cm	2.69	(1.2-6.3)	3.21	(1.4-7.6)	5.54	(1.7-18.4)
FIGO (stage IV vs III)	1.76	(1.3-2.4)	1.70	(1.2-2.5)	1.64	(0.9-2.9)
Karnofsky index	0.77	(0.7-0.9)	0.89	(0.8-1.0)	1.07	(0.8-1.4)

TABLE 7 Estimated hazard ratios (HR) and 95 % confidence intervals (CI) in landmark models (with 2-year windows) for the Ovarian cancer data

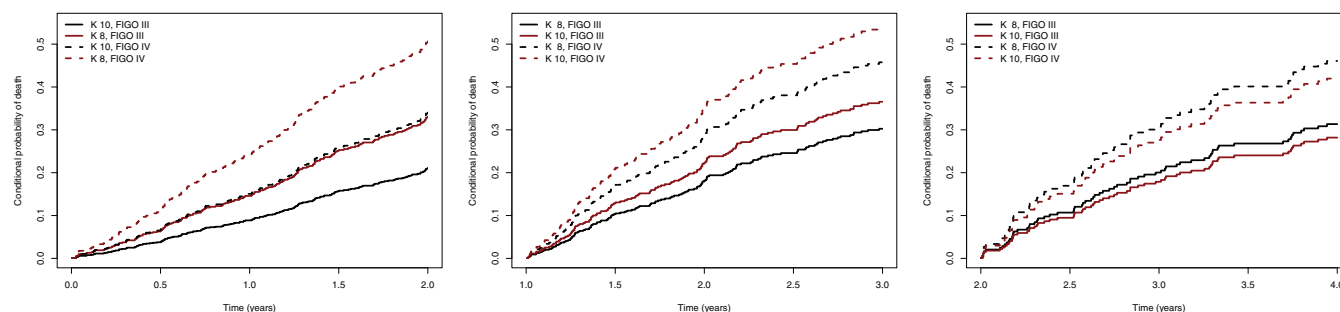


FIGURE 12 The conditional probability of dying for patients with diameter <1 cm with respect to stage and two chosen levels of Karnofsky index. Two-year conditional probabilities for patients still at risk at the beginning of each time window are estimated [Colour figure can be viewed at wileyonlinelibrary.com]

Prediction: The hazard ratios describe the relative effect of each covariate, but do not tell anything about the absolute risk of the patients. To this end, Figure 12 shows risk estimates for a set of covariate values. The probability of dying in the first 2 years is comparable in size to the conditional probability of dying in the next 2 years at each of chosen time points. As seen in Table 7, the Karnofsky index at baseline is crucial for the prognosis in the first 2 years, but less relevant for patients who survive the initial period. Obviously, observation of an updated Karnofsky index could change this conclusion.

In conclusion, the fitted Cox regression model enabled estimation of the absolute risk of dying within 2 years, even in the presence of a covariate for which the PH assumption was not satisfied. For that purpose, the technique of landmarking proved very useful.

7 | SUMMARY AND DISCUSSION

In general multistate models, the *intensity* is the basic parameter⁸ and we have argued that, in the analysis of time-to-event data from observational studies, the intensity is, therefore, an obvious parameter to target. Focus has been on a single occurrence of a single type of event, such as (cause-specific) death, onset/diagnosis of a disease, or first hospital re-admission. Recurrent themes have been that hazard models known from survival analysis are applicable in such situations and that studies of this kind have a number of common features. These include, for example, specification of the time axis for analysis, how to deal with incomplete observation in the form of right-censoring and delayed entry, and how to use and interpret models including time-dependent covariates. Also, the concept of immortal time bias is relevant in all such studies.

We have provided some checklists that we find useful to consider, however, it is important to emphasize that these checklists cannot be taken as “cook books” on how to conduct time to event analysis in observational studies. Rather, they are meant as guidelines and we have also emphasized that the most important item to consider when planning such an analysis is to clearly specify the research question and think about to what extent the available data allow an answer to that question. We have also identified research questions for which an intensity model only provides one step toward an answer and where further analyses are needed. These include risk prediction for nonfatal events and causal inference.

Finally, we have presented some worked examples using the methods summarized in earlier sections and going through the checklists provided. Our examples illustrate also the need to interpret the results with some caution, taking into account both the limitations of the data at hand and the underlying assumptions, which should be carefully checked, possibly triggering some additional analysis. Further details concerning these examples are collected as Appendix S1 that also includes information on how the analysis results were obtained using R.

Even though the paper is not short, it fails to discuss a number of aspects that are also of importance. These include most mathematical details about properties of the methods, as well as more comprehensive analyses of data with competing risks, recurrent events, and more general multistate models.^{9,66} We have focussed on the Cox regression model throughout (and to a lesser extent the piecewise exponential/‘Poisson’ model) and discussion of AFT models, additive hazards models as well as random effects (‘frailty’) models, for example, joint models for the event intensity and an internal time-dependent covariate, is not included.^{52,67} The same holds for models for relative survival⁶⁸ and how to deal with interval-censoring.¹⁶ Some of these may be topics for forthcoming papers from the STRATOS TG8 topic group.

ACKNOWLEDGEMENTS

Michal Abrahamowicz is a James McGill Professor at McGill University. His research is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) grant 228203 and the Canadian Institutes of Health Research (CIHR) grant PJT-148946. The research of Maja Pohar Perme is supported by Slovenian Research Agency (grant P3-0154, “Methodology for data analysis in medical sciences”). Richard J. Cook is Faculty of Mathematics Research Chair. His research is supported by grants from the Natural Sciences and Engineering Research Council of Canada (RGPIN 155849 and RGPIN 04207) and the Canadian Institutes of Health Research (FRN 13887). Jeremy M. Taylor is funded by NIH grant CA-46592.

DATA AVAILABILITY STATEMENT

The data are provided in Appendix S1.

ORCID

Per Kragh Andersen  <https://orcid.org/0000-0003-0490-0885>

Maja Pohar Perme  <https://orcid.org/0000-0002-3412-2642>

Richard J. Cook  <https://orcid.org/0000-0002-1414-4908>

Michal Abrahamowicz  <https://orcid.org/0000-0002-3172-3952>

REFERENCES

- Altman DG, De Stavola BL, Love SB, Stepniowska KA. Review of survival analyses published in cancer journals. *Br J Cancer*. 1995;72(2):511-518.
- Sauerbrei W, Abrahamowicz M, Altman DG, le Cessie S, Carpenter J, STRATOS Initiative. STREngthening analytical thinking for observational studies: the STRATOS initiative. *Stat Med*. 2014;33(30):5413-5432. <https://doi.org/10.1002/sim.6265>.
- Kessing LV, Søndergaard L, Forman JL, Andersen PK. Lithium treatment and risk of dementia. *Arch Gen Psychiatry*. 2008;65:1331-1335.
- Suissa S. Immortal time bias in pharmacoepidemiology. *Am J Epidemiol*. 2007;167(4):492-499.
- Anderson JR, Cain KC, Gelber RD. Analysis of survival by tumor response. *J Clin Oncol*. 1983;1(11):710-719.
- Keiding N. Delayed entry. *Encyclopedia of Biostatistics*. Hoboken, NJ: John Wiley & Sons, Ltd; 2005.
- Andersen PK, Geskus RB, Witte DT, Putter H. Competing risks in epidemiology: possibilities and pitfalls. *Int J Epidemiol*. 2012;41(3):861-870.
- Andersen PK, Borgan Ø, Gill RD, Keiding N. *Statistical Models Based on Counting Processes*. New York, NY: Springer; 1993.
- Cook RJ, Lawless JF. *The Statistical Analysis of Recurrent Events*. New York, NY: Springer; 2007.
- Cox DR. Regression models and life-tables. *J Royal Stat Soc Ser B (Methodol)*. 1972;34(2):187-220.
- Breslow NE. Covariance analysis of censored survival data. *Biometrics*. 1974;30:89-99.
- Kalbfleisch JD, Prentice RL. *The Statistical Analysis of Failure Time Data*. Hoboken, NJ: John Wiley & Sons, Inc; 2002.
- Cox DR. Partial likelihood. *Biometrika*. 1975;62:269-276.
- Andersen PK, Gill RD. Cox's regression model for counting processes: a large sample study. *Ann Stat*. 1982;10(4):1100-1120.
- Clayton DG, Hills M. *Statistical Models in Epidemiology*. Oxford, UK: Oxford University Press; 1993.
- Joly P, Commenges D, Letenneur L. A penalized likelihood approach for arbitrarily censored and truncated data: application to age-specific incidence of dementia. *Biometrics*. 1998;54:185-194.
- Royston P, Lambert PC. *Flexible Parametric Survival Analysis Using Stata: Beyond the Cox Model*. College Station, TX: Stata Press; 2011.
- Aalen OO. A linear regression model for the analysis of life times. *Stat Med*. 1989;8(8):907-925. <https://doi.org/10.1002/sim.4780080803>.
- Lin DY, Ying Z. Semiparametric analysis of the additive risk model. *Biometrika*. 1994;81(1):61-71. <https://doi.org/10.1093/biomet/81.1.61>.

20. Martinussen T, Scheike TH. *Dynamic Regression Models for Survival Data*. New York, NY: Springer; 2007.
21. Aalen O, Borgan Ø, Gjessing H. *Survival and Event History Analysis: A Process Point of View*. New York, NY: Springer; 2008.
22. Wei LJ. The accelerated failure time model: a useful alternative to the Cox regression model in survival analysis. *Stat Med*. 1992;11:1871-1879.
23. Keiding N, Andersen PK, Klein JP. The role of frailty models and accelerated failure time models in describing heterogeneity due to omitted covariates. *Stat Med*. 1997;16(2):215-224. [https://doi.org/10.1002/\(SICI\)1097-0258\(19970130\)16:2<215::AID-SIM481>3.0.CO;2-J](https://doi.org/10.1002/(SICI)1097-0258(19970130)16:2<215::AID-SIM481>3.0.CO;2-J).
24. Bycott P, Taylor JMG. A comparison of smoothing techniques for CD4 data measured with error in a time-dependent Cox proportional hazards model. *Stat Med*. 1998;17(18):2061-2077.
25. Andersen PK, Liestøl K. Attenuation caused by infrequently updated covariates in survival analysis. *Biostatistics*. 2003;4(4):633-649.
26. de Bruijne MH, le Cessie S, Kluin-Nelemans HC, van Houwelingen HC. On the use of Cox regression in the presence of an irregularly observed time-dependent covariate. *Stat Med*. 2001;20(24):3817-3829.
27. Wulfsohn MS, Tsiatis AA. A joint model for survival and longitudinal data measured with error. *Biometrics*. 1997;53:330-339.
28. Tsiatis AA, Davidian M. Joint modeling of longitudinal and time-to-event data: an overview. *Stat Sin*. 2004;4(3):809-834.
29. Horwitz RI, Feinstein AR. The problem of "protopathic bias" in case-control studies. *Am J Med*. 1980;68(2):255-258.
30. Gasparrini A. Modeling exposure-lag-response associations with distributed lag non-linear models. *Stat Med*. 2014;33(5):881-899.
31. Sylvestre MP, Abrahamowicz M. Flexible modeling of the cumulative effects of time-dependent exposures on the hazard. *Stat Med*. 2009;28(27):3437-3453.
32. Therneau TM, Grambsch PM. *Modeling Survival Data: Extending the Cox Model*. New York, NY: Springer; 2000.
33. Lin DY, Wei LJ, Ying Z. Checking the Cox model with cumulative sums of martingale-based residuals. *Biometrika*. 1993;80(3):557-572. <https://doi.org/10.1093/biomet/80.3.557>.
34. Abrahamowicz M, MacKenzie TA. Joint estimation of time-dependent and non-linear effects of continuous covariates on survival. *Stat Med*. 2007;26:392-408.
35. Wynant W, Abrahamowicz M. Impact of the model-building strategy on inference about nonlinear and time-dependent covariate effects in survival analysis. *Stat Med*. 2014;33(19):3318-3337. <https://doi.org/10.1002/sim.6178>.
36. Schmoor C, Schumacher M. Effects of covariate omission and categorization when analysing randomized trials with the Cox model. *Stat Med*. 1997;16(3):225-237.
37. Bretagnolle J, Huber-Carol C. Effects of omitting covariates in Cox's model for survival data. *Scand J Stat*. 1988;15:125-138.
38. Struthers CA, Kalbfleisch JD. Misspecified proportional hazards models. *Biometrika*. 1986;74(2):363-369.
39. Hernán MA. The hazards of hazard ratios. *Epidemiology*. 2010;21(1):13-15.
40. Aalen OO, Cook RJ, Røysland K. Does Cox analysis of a randomized survival study yield a causal treatment effect? *Lifetime Data Anal*. 2015;21(4):579-593.
41. Martinussen T, Vansteelandt S, Andersen PK. Subtleties in the interpretation of hazard ratios. *Lifetime Data Anal*. 2020;26:833-855. <https://doi.org/10.1007/s10985-020-09501-5>.
42. Crowley JJ, Hu M. Covariance analysis of heart transplant survival data. *J Am Stat Assoc*. 1977;72:27-36.
43. Clark DA, Stinson EB, Griep RB, Schroeder JS, Shumway NE, Harrison DC. Cardiac transplantation in man. *Ann Intern Med*. 1971;75(1):15-21.
44. Redmond C, Fisher B, Wieand HS. The methodologic dilemma in retrospectively correlating the amount of chemotherapy received in adjuvant therapy protocols with disease-free survival. *Cancer Treat Rep*. 1983;67:519-526.
45. Bonadonna G, Valagussa P. Dose-response effect of adjuvant chemotherapy in breast cancer. *N Engl J Med*. 1981;304(1):10-15.
46. Van Houwelingen HC, Putter H. Comparison of stopped Cox regression with direct methods such as pseudo-values and binomial regression. *Lifetime Data Anal*. 2015;21(2):180-196.
47. Andersen PK, Perme MP. Pseudo-observations in survival analysis. *Stat Methods Med Res*. 2010;19(1):71-99.
48. Van Houwelingen HC. Dynamic prediction by landmarking in event history analysis. *Scand J Stat*. 2007;34(1):70-85.
49. Jewell NP, Nielsen JP. A framework for consistent prediction rules based on markers. *Biometrika*. 1993;80(1):153-164.
50. Suresh K, Taylor JMG, Spratt DE, Daignault S, Tsodikov A. Comparison of joint modeling and landmarking for dynamic prediction under an illness-death model. *Biom J*. 2017;59(6):1277-1300.
51. Keogh RH, Seaman SR, Barrett JK, Taylor-Robinson D, Szczesniak R. Dynamic prediction of survival in cystic fibrosis: a landmarking analysis using UK patient registry data. *Epidemiology*. 2019;30(1):29-37.
52. Rizopoulos D. Dynamic predictions and prospective accuracy in joint models for longitudinal and time-to-event data. *Biometrics*. 2011;67(3):819-829.
53. Fine JP, Gray RJA. Proportional hazards model for the subdistribution of a competing risk. *J Am Stat Assoc*. 1999;94(446):496-509. <https://doi.org/10.1080/01621459.1999.10474144>.
54. Rubin DB. Causal inference using potential outcomes: design, modeling. *Decis J Am Stat Assoc*. 2005;100(469):322-331.
55. Goetghebuer E, le Cessie S, De Stavola B, Moodie E, Waernbaum I. Formulating causal questions and principled statistical answers. *Statistics in Medicine*. 2020. <http://dx.doi.org/10.1002/sim.8741>.
56. Hernán MA, Robins JM. *Causal Inference: What If*. Boca Raton, FL: Chapman & Hall/CRC Press; 2020.
57. Chen PY, Tsiatis AA. Causal inference on the difference of the restricted mean lifetime between two groups. *Biometrics*. 2001;57(4):1030-1038.
58. Daniel RM, Cousens S, De Stavola B, Kenward MG, Sterne J. Methods for dealing with time-dependent confounding. *Stat Med*. 2013;32(9):1584-1618.

59. Blinc A, Kozak M, Šabovič M, et al. Survival and event-free survival of patients with peripheral arterial disease undergoing prevention of cardiovascular disease. *Int Angiol*. 2017;35:216-227.
60. Blinc A, Kozak M, Šabovič M, et al. Prevention of ischemic events in patients with peripheral arterial disease – design, baseline characteristics and 2-year results an observational study. *Int Angiol*. 2011;30:555-566.
61. Puri P, Sanyal AJ. Nonalcoholic fatty liver disease: definitions, risk factors and workup. *Clin Liver Dis*. 2012;1:99-103.
62. Tapper EB, Loomba R. Nonalcoholic fatty liver disease, metabolic syndrome, and the fight that will define clinical practice for a generation of hepatologists. *Hepatology*. 2018;67:1657-1659.
63. Allen AM, Therneau TM, Larson JJ, Coward A, Somers VK, Kamath PS. Nonalcoholic fatty liver disease incidence and impact on metabolic burden and death: a 20 year community study. *Hepatology*. 2018;67:1726-1736.
64. St Sauver JL, Grossardt BR, Yawn BP, et al. Data resource profile: the Rochester epidemiology project (REP) medical records-linkage system. *Int J Epidemiol*. 2012;41(6):1614-1624. <https://doi.org/10.1093/ije/dys195>.
65. Van Houwelingen HC, Putter H. *Dynamic Prediction in Clinical Survival Analysis*. Boca Raton, FL: CRC Press; 2012.
66. Cook RJ, Lawless JF. *Multistate Models for the Analysis of Life History Data*. Boca Raton, FL: Chapman & Hall/CRC Press; 2018.
67. Hougaard P. *Analysis of Multivariate Survival Data*. New York, NY: Springer; 2000.
68. Perme MP, Stare J, Estève J. On estimation in relative survival. *Biometrics*. 2012;68(1):113-120.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

How to cite this article: Kragh Andersen P, Pohar Perme M, van Houwelingen HC, et al. Analysis of time-to-event for observational studies: Guidance to the use of intensity models. *Statistics in Medicine*. 2021;40:185–211. <https://doi.org/10.1002/sim.8757>